Week 8, July-Sept 2017
MOOCS Decision Modeling Course

Networks Models

NPTEL

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Module 36

Fundamental Matrices and Network Simplex Method

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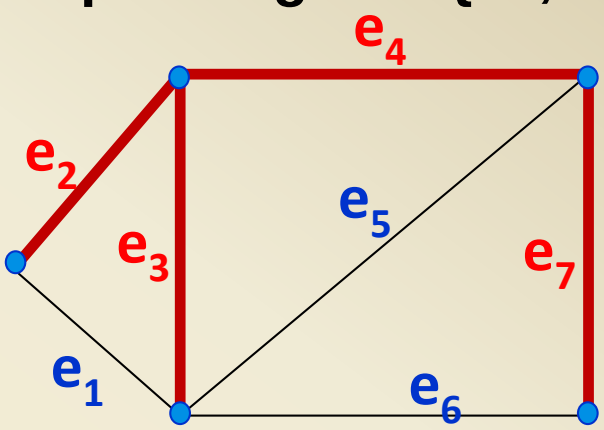
Problem on Fundamental Matrix

Consider the graph. With respect to the spanning tree {e2, e3, e4, e7}, find out the following:



- **Fundamental Circuit matrix (B_f)**
- **Fundamental Cut-set matrix (C_f)**
- **Show that the matrices B_f and C_f are orthogonal to each other.**

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Fundamental Circuit Matrix

Consider the graph. With respect to the spanning tree {e2, e3, e4, e7}, find out the following:

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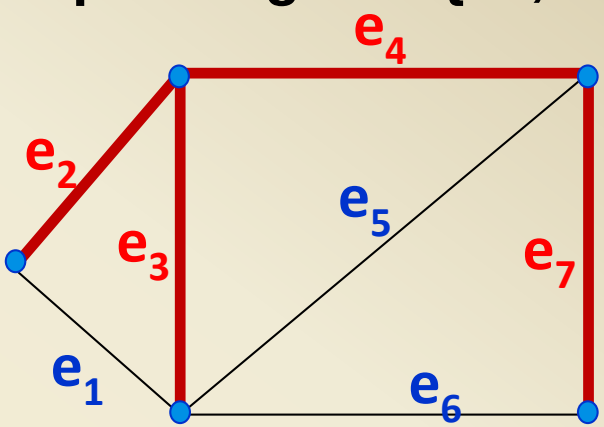
- **Fundamental Cut-set matrix (C_f)**
- **Show that the matrices B_f and C_f are orthogonal to each other.**



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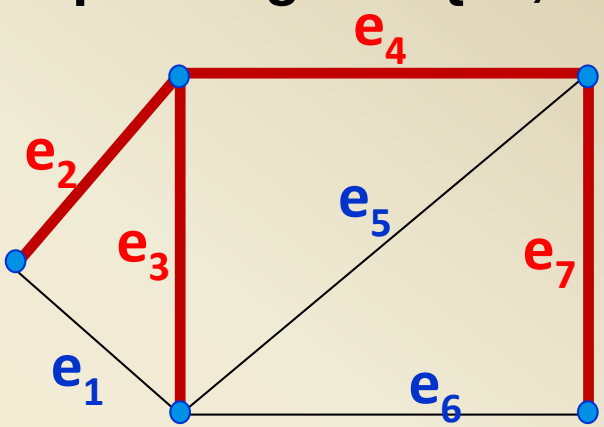
Fundamental Circuit matrix (B_f)

	e_2	e_3	e_4	e_7	e_1	e_5	e_6
B_1	1	1	0	0	1	0	0
B_2	0	1	1	0	0	1	0
B_3	0	1	1	1	0	0	1

Fundamental Cutset Matrix

Consider the graph. With respect to the spanning tree {e2, e3, e4, e7}, find out the following:

- **Fundamental Circuit matrix (B_f)**
- **Fundamental Cut-set matrix (C_f)**
- **Show that the matrices B_f and C_f are orthogonal to each other.**



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Fundamental Cutset matrix (C_f)

	e_2	e_3	e_4	e_7	e_1	e_5	e_6
C_1	1	0	0	0	1	0	0
C_2	0	1	0	0	1	1	1
C_3	0	0	1	0	0	1	1
C_4	0	0	0	1	0	0	1

B_f and C_f are Orthogonal

Fundamental Circuit matrix (B_f)

B_f	e_2	e_3	e_4	e_7	e_1	e_5	e_6
B_1	1	1	0	0	1	0	0
B_2	0	1	1	0	0	1	0
B_3	0	1	1	1	0	0	1

Fundamental Cut-set matrix (C_f)

C_f	e_2	e_3	e_4	e_7	e_1	e_5	e_6
C_1	1	0	0	0	1	0	0
C_2	0	1	0	0	1	1	1
C_3	0	0	1	0	0	1	1
C_4	0	0	0	1	0	0	1

Show that the matrices B_f and C_f are orthogonal to each other.

$B_f * C_f$ transposed = Φ mod 2

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B_f	e_1	e_2	e_3	e_4	e_5	e_6	e_7
B_1	1	1	1	0	0	0	0
B_2	0	0	1	1	1	0	0
B_3	0	0	1	1	0	1	1

*

C_f T	C_1	C_2	C_3	C_4
e_1	1	1	0	0
e_2	1	0	0	0
e_3	0	1	0	0
e_4	0	0	1	0
e_5	0	1	1	0
e_6	0	1	1	1
e_7	0	0	0	1

=

	C_1	C_2	C_3	C_4
B_1	2	2	0	0
B_2	0	2	2	0
B_3	0	2	2	2

= Φ mod 2

Network Simplex Method

Consider the Transportation Problem given below. Find its optimal solution using the Network Simplex Method.

	Demand 1	Demand 2	Availability (a_i)
Supply 1	2/unit	2/unit	20
Supply 2	1/unit	2/unit	60
Demand (d_j)	40	40	Total=80

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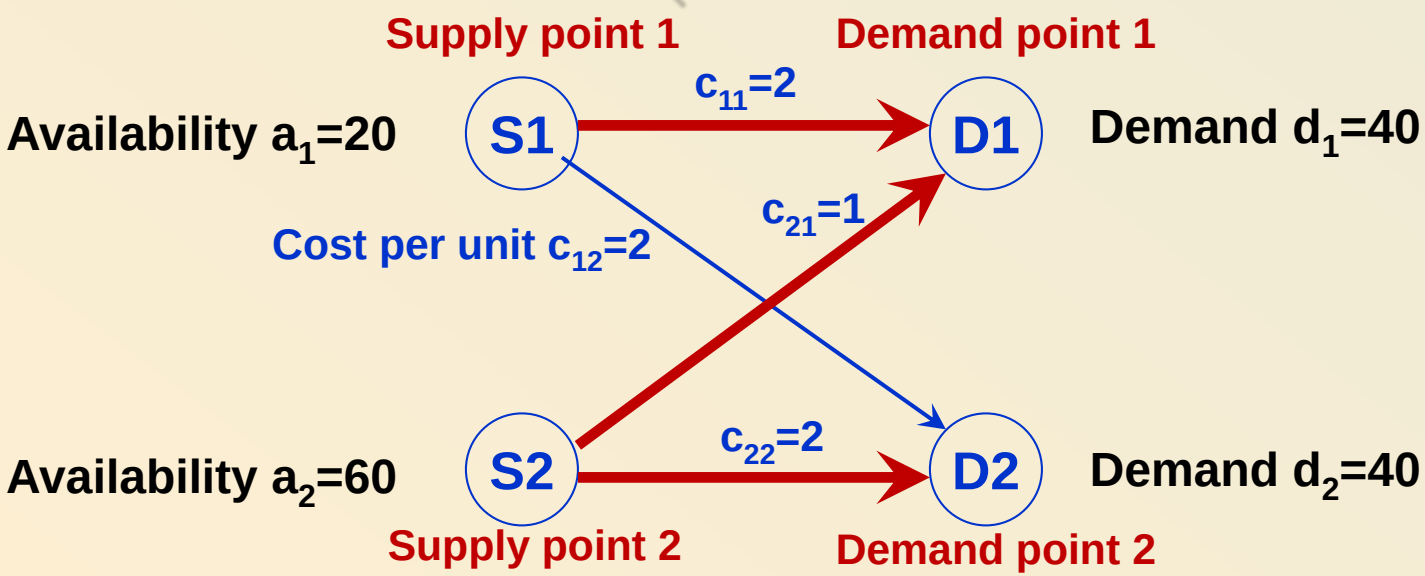
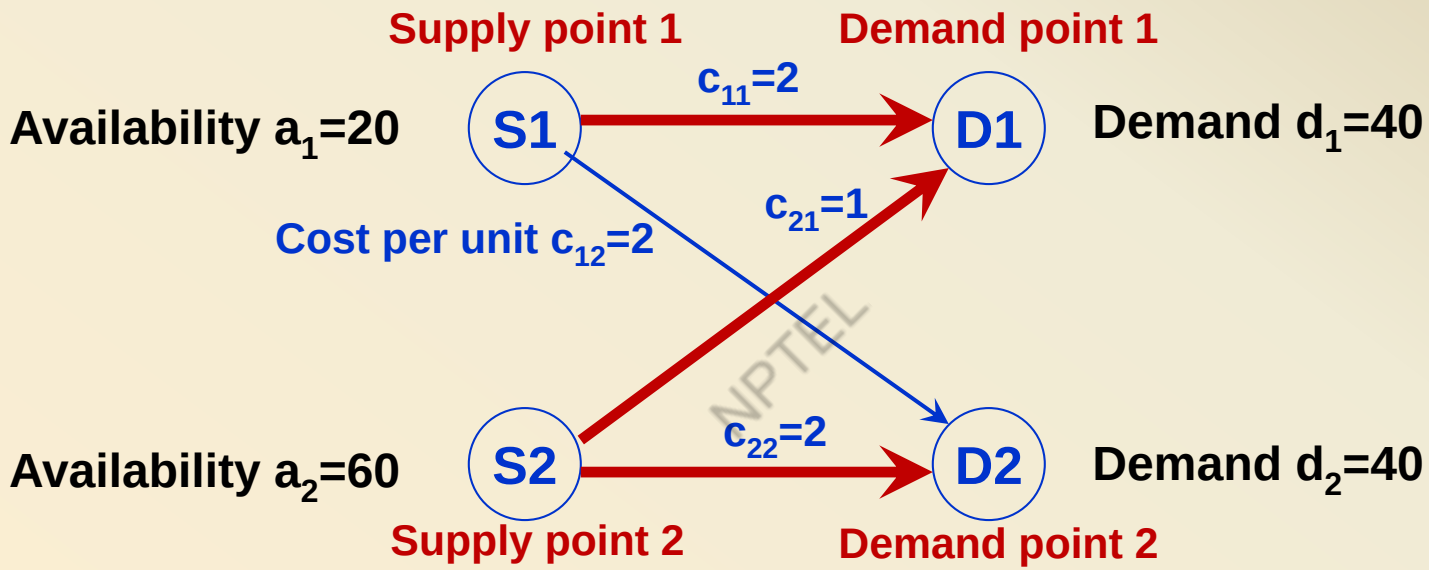


Network Simplex Method

Transportation Problem



- Each spanning tree of the graph is a candidate solution of the transportation problem using network simplex method.
- Let the starting solution be a Spanning tree {S1-D1, S2-D1, S2-D2}.

Network Simplex Method

The Network Simplex method is described as follows:

1) ***Start with an initial solution (a reasonably good one) in the form of a Spanning Tree.***
2) ***Find flows through the arcs by flow balance.***
3) ***Find Node Potentials of all the nodes in the graph using costs per unit pertaining to the Spanning tree.***

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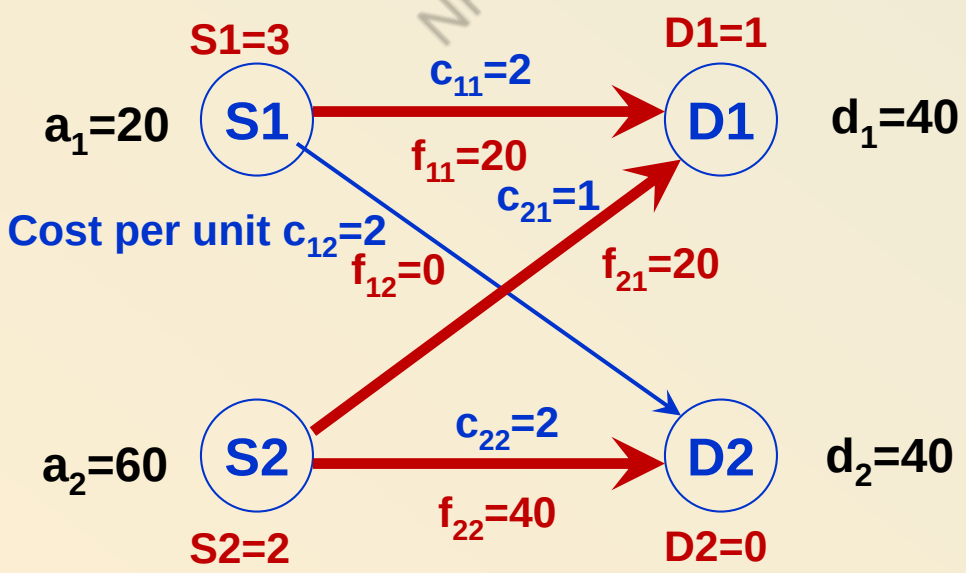

Network Simplex Method

4) Find $(z_{ij} - c_{ij})$ values for the arcs other than the Spanning tree. If all the values are negative, the flow pattern is optimal, else, identify the arc with most positive $(z_{ij} - c_{ij})$ value – this is the entering arc.

5) Find a fundamental circuit with this arc selected in 4). Find flow change Δ increasing flow in entering arc. A spanning tree arc with zero flow will be the leaving arc.

6) Now a new spanning tree is obtained. Repeat steps 2) to 4) till the flow pattern in optimal.

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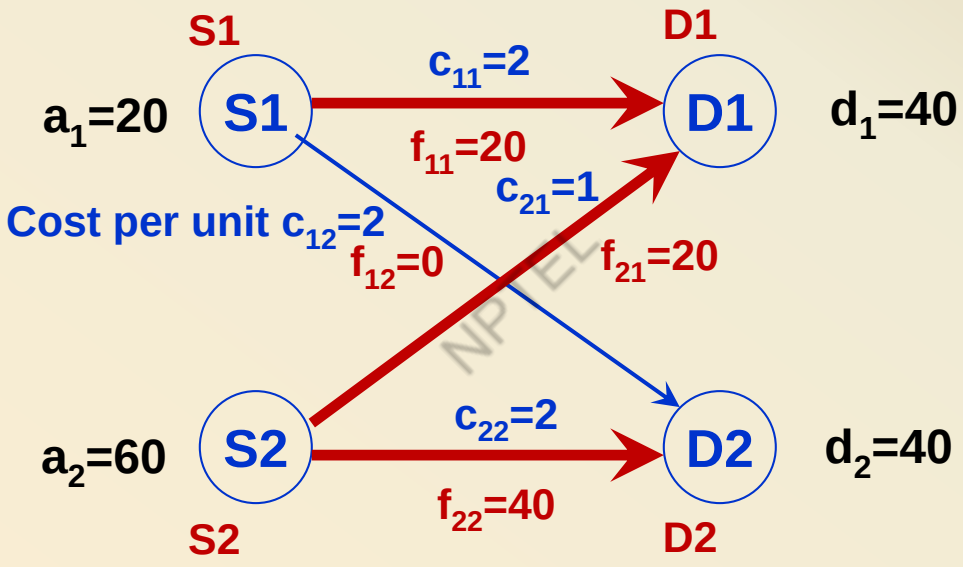
Flow Balance

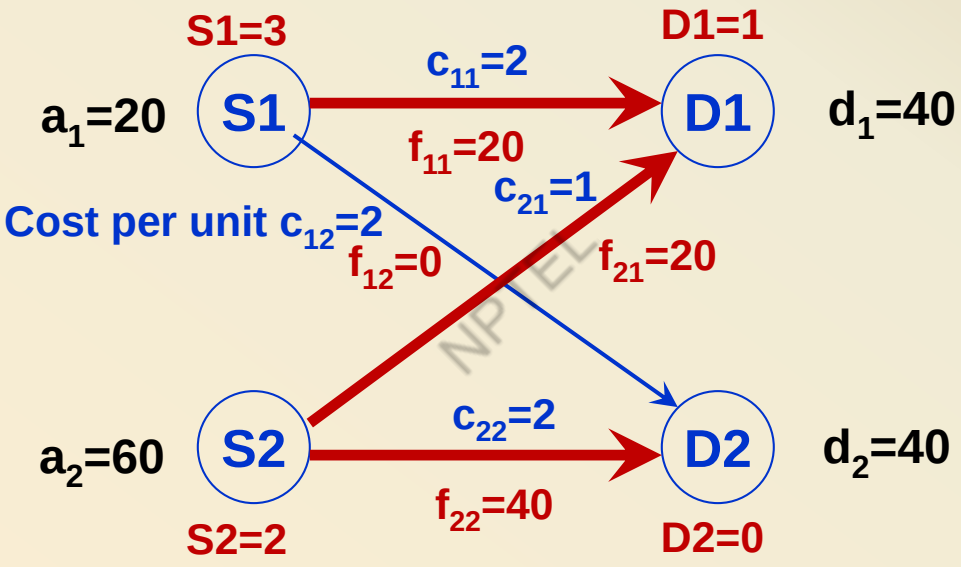
Balancing the flows, the maximal flows pertaining to the initial solution are indicated.



Starting solution is the Spanning tree {S1-D1, S2-D1, S2-D2}.

Node Potentials

Node Potentials of all the nodes in the graph are found using costs per unit pertaining to the Spanning tree.



For the Spanning tree arcs:

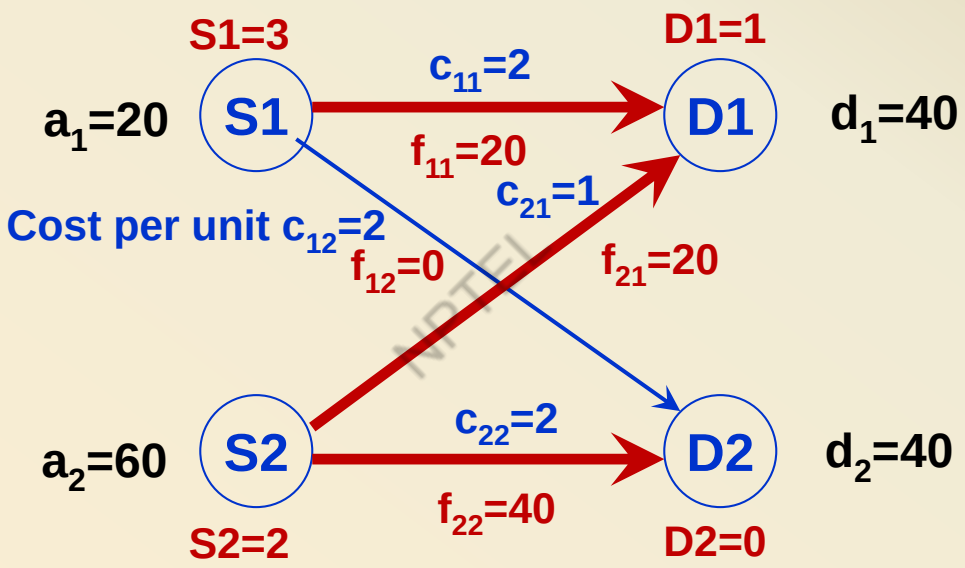
S1 – D1 = 2; S2 – D1 = 1; S2 – D2 = 2. **Put D2 = 0 (say)**

Hence, Node Potentials: S2 = 2; D1 = 1; S1 = 3

$(z_{ij} - c_{ij})$ Values

$(z_{ij} - c_{ij})$ values for the arcs other than the Spanning tree are computed.

Note: z_{ij} is Difference in Node Potential from i to j



Non-spanning tree arc: 14

For this arc: $(z_{ij} - c_{ij})$ = (S1 – D2) – 2 = (3 – 0) – 2 = 1 Positive!
Hence, this arc is an entering arc.

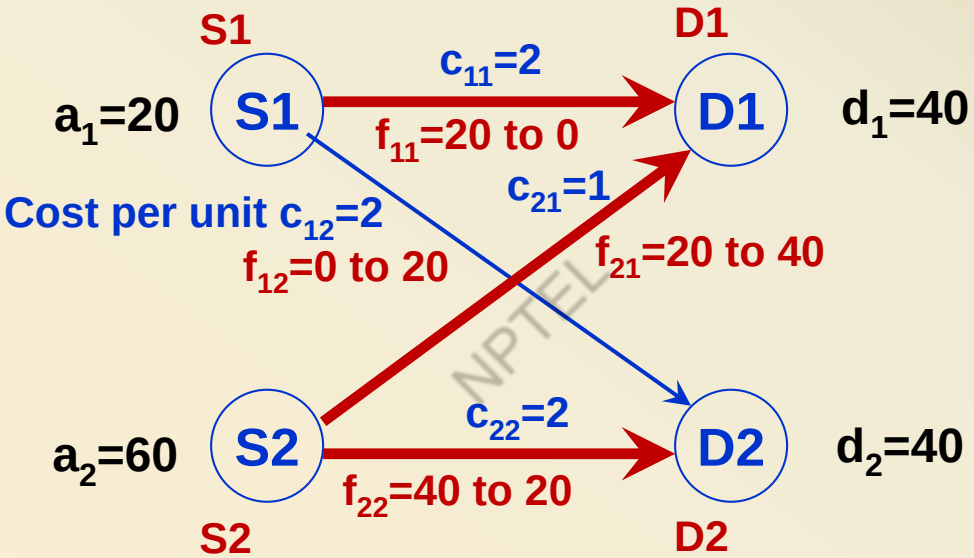
Fundamental Circuit: {S1-D1, S2-D1, S2-D2, S1-D2}.

Flow Change Δ

Flow change Δ is found out increasing flow in the entering arc. A spanning tree arc with zero flow will be the leaving arc.

Entering Arc: S1-D2

Leaving Arc: S1-D1



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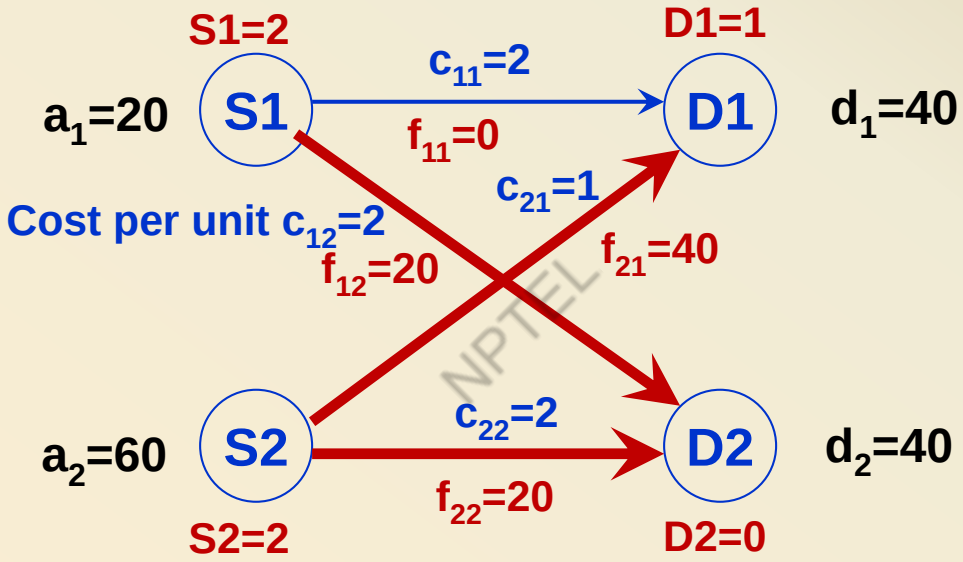
Flow Change: As flow in entering arc S1-D2 goes up, flow in S1-D1 goes down, S2-D1 goes up, S2-D2 goes down.

Note the flow change possible. Δ = 20 (Why?)

Hence, new flow values are indicated in the diagram. New spanning tree is {S1-D2, S2-D1, S2-D2}.

New Spanning Tree

New spanning tree is {S1-D2, S2-D1, S2-D2}. Node Potentials are obtained. ($z_{ij} - c_{ij}$) value for non-spanning tree arc is computed.



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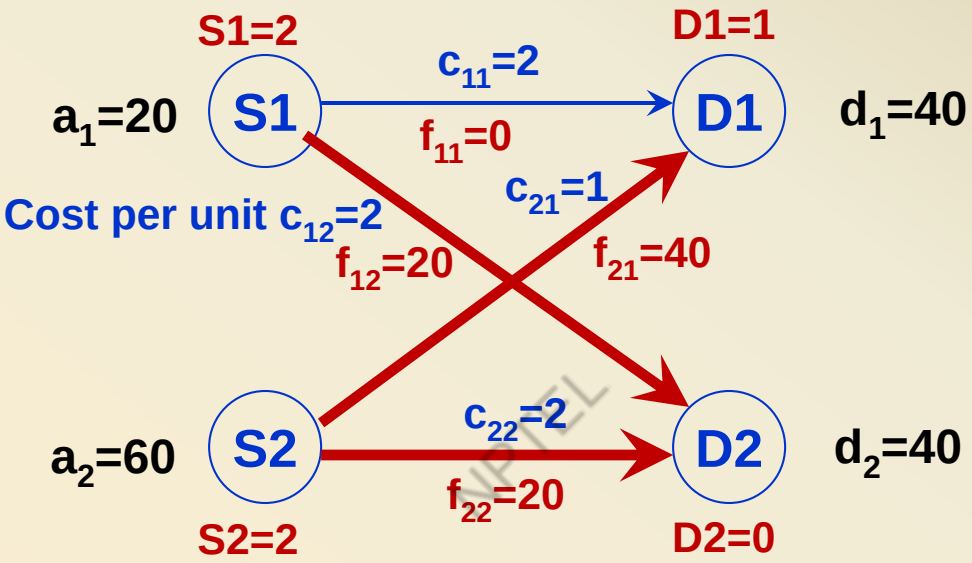
New Node Potential Values are indicated. Why are the values as indicated?

Non-spanning tree arc: 11

For this arc: ($z_{ij} - c_{ij}$) = (S1 – D1) – 2 = (2 – 1) – 2 = –1 Negative!
Hence, the current flow pattern is optimal.

Optimal Solution



	Demand 1	Demand 2	Availability
Supply 1	No Flow	Flow = 20	20
Supply 2	Flow = 40	Flow = 40	60
Demand	40	40	Total=80

Total Cost = (20*2)+(40*1)+(20*2) = 120

Another Problem

Consider the Transportation Problem given below. Find its optimal solution using the Network Simplex Method.

	Demand 1	Demand 2	Demand 3	Availability (a_i)
Supply 1	2/unit	2/unit	1/unit	20
Supply 2	1/unit	2/unit	1/unit	60
Demand (d_j)	30	30	20	Total=22

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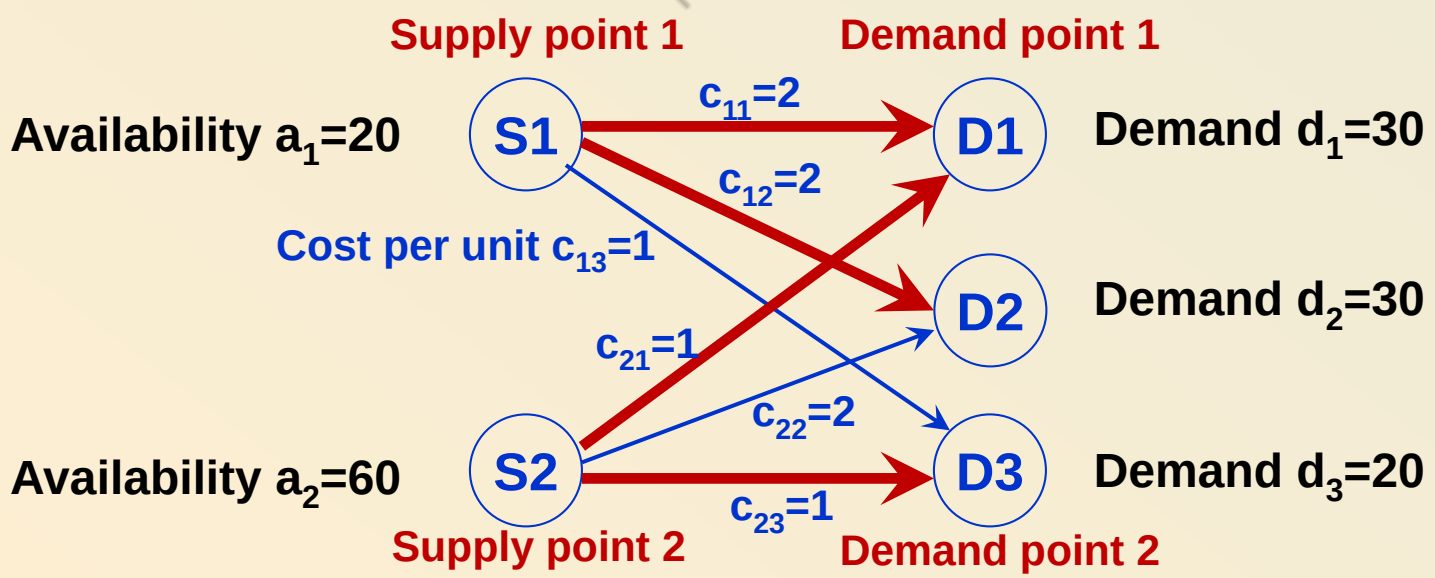

Module 37

Maximal Flow Problems

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Flow Network

- A **flow network** is a simple connected weighted and directed graph.
- Also known as a **transport network**.
- A flow network is shown below.
- The flow is from **source s to sink t**.
- Each arc has a **forward** and **reverse** flow **capacity**.

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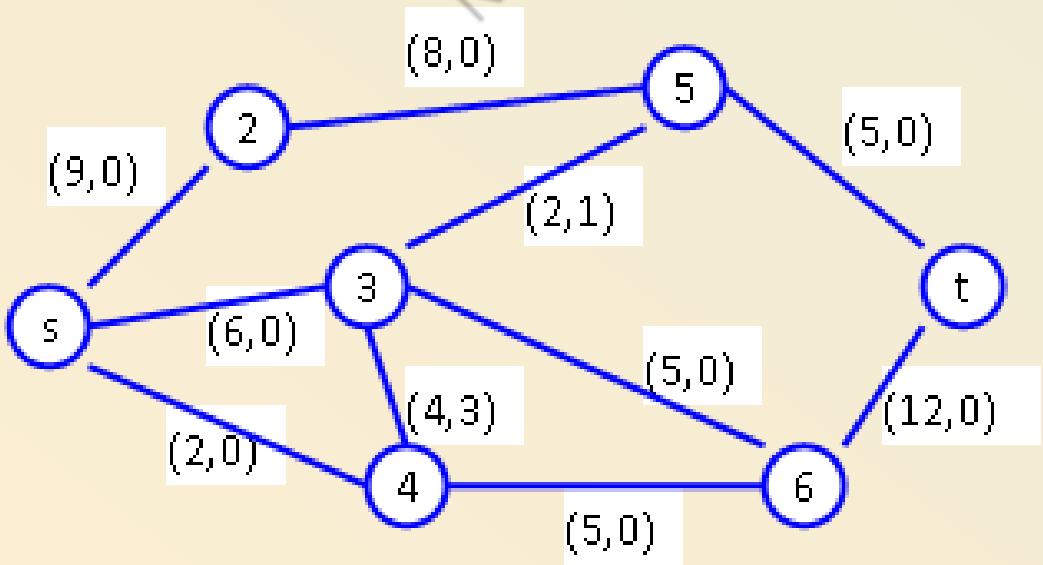

Flow Network

- A **flow network** is a simple connected weighted and directed graph.
- **Assumptions** include:
  - Single source and single sink
  - infinite node capacities
  - lossless flow for arcs
  - No minimum limit for flow

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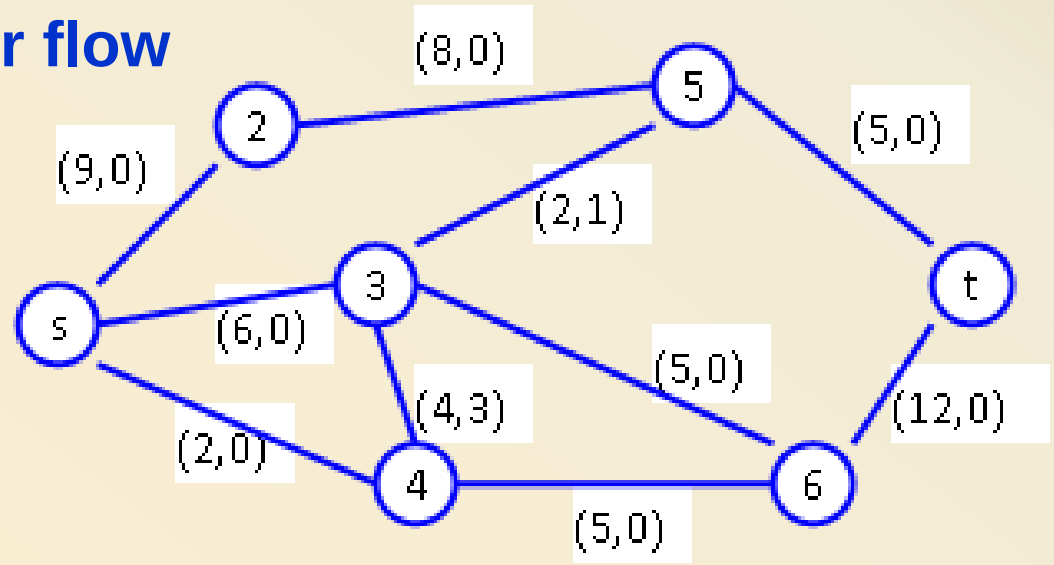

Network Flow Assumptions

1) Single source and single sink

With multiple sources and multiple sinks, connect all the sources to a super-source and all the sinks to a super-sink with arcs of infinite flow capacity

2) Infinite node capacities

With limited node capacities, replace the node with two nodes – connect incoming arcs to first and outgoing arcs to second. Join the nodes with an arc of limited node capacity specified.

3) Lossless flow for arcs

With flow losses in arcs, use Linear Programming formulation.

4) No minimum limit for flow

With minimum and maximum flow limits, use an appropriate network flow algorithm.

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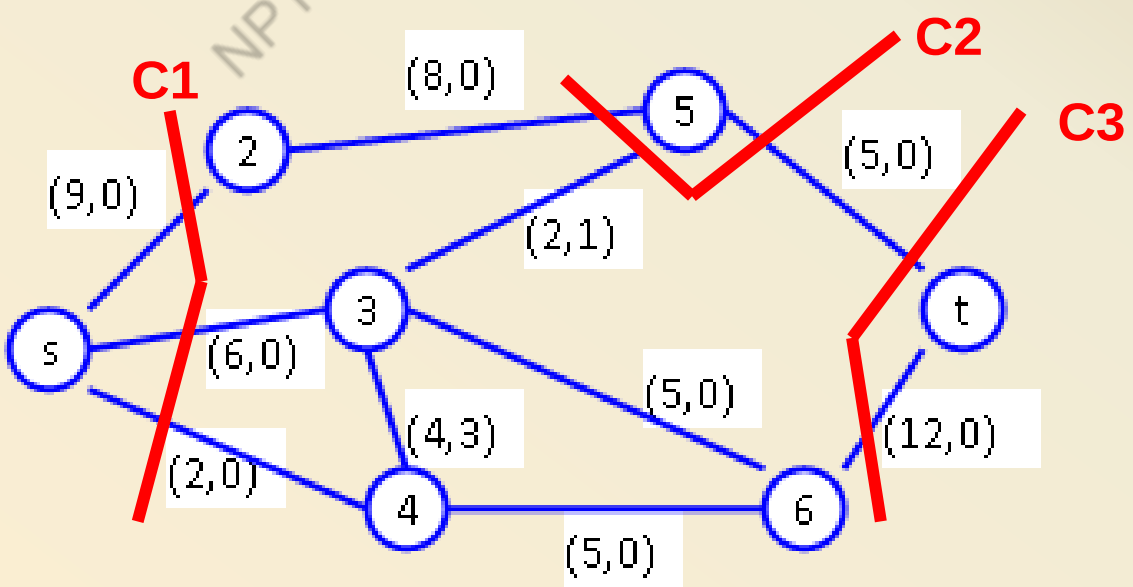
Cut and Cutset

- In a flow network, a cut is a cutset that separates the source from the sink.
- A cutset is a minimal set of edges whose removal leaves the graph disconnected
- A cut also disconnects the flow network with source and sink in different components.

- Consider the 3 cutsets C1, C2, C3.
- Which one is not a cut?



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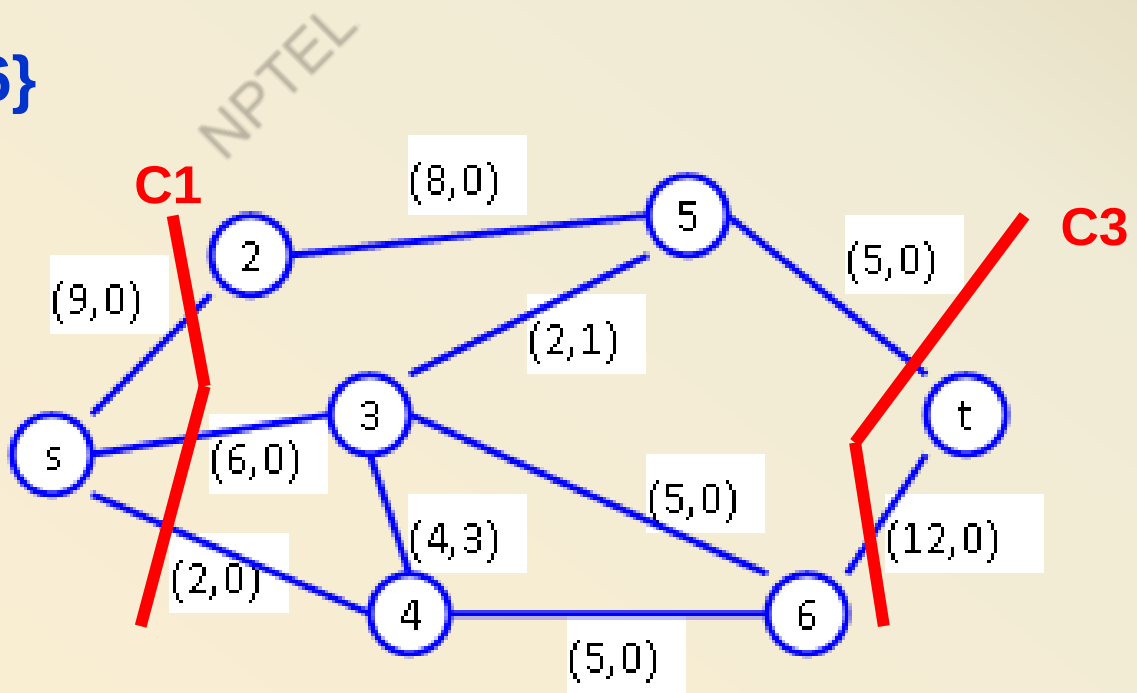
Cut and Cutset

- A cut is expressed in terms of 2 sets of nodes: p and $\bar{p}$
- The cut partitions the nodes into two sets p and $\bar{p}$
- In the figure below, for cut C1,

  p is {s} and $\bar{p}$ is {2,3,4,5,6,t}
- For cut C3,

  p is {s,2,3,4,5,6}

  and $\bar{p}$ is {t}



Capacity of a Cut

- Capacity of a cut is shown as $C(p, \overline{p})$

Cut	Shown as	$C(p, \overline{p})$
C1	{s}	17
C2	{s,2}	16
C3	{s,2,3,4,5,6}	17

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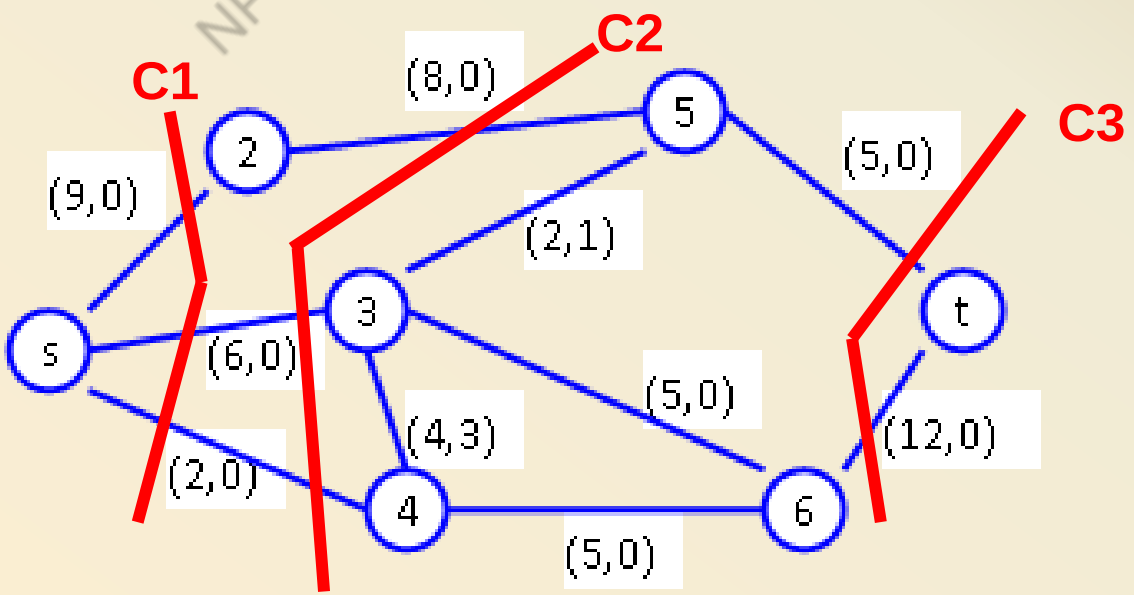
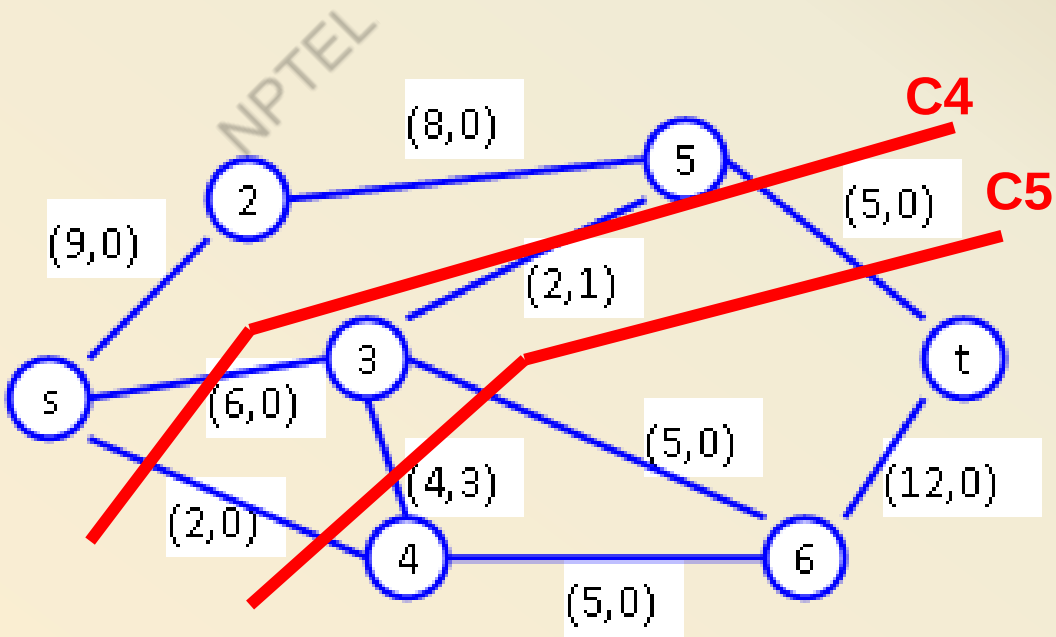
Note: Always look for source to sink flow through the cut.



Capacity of a Cut

- Capacity of a cut is shown as $C(p, \overline{p})$
- Find the capacity of the following cuts:
- C4: {s,2,5} and C5: {s,2, 3, 5}

Note: Always look for source to sink flow through the cut.

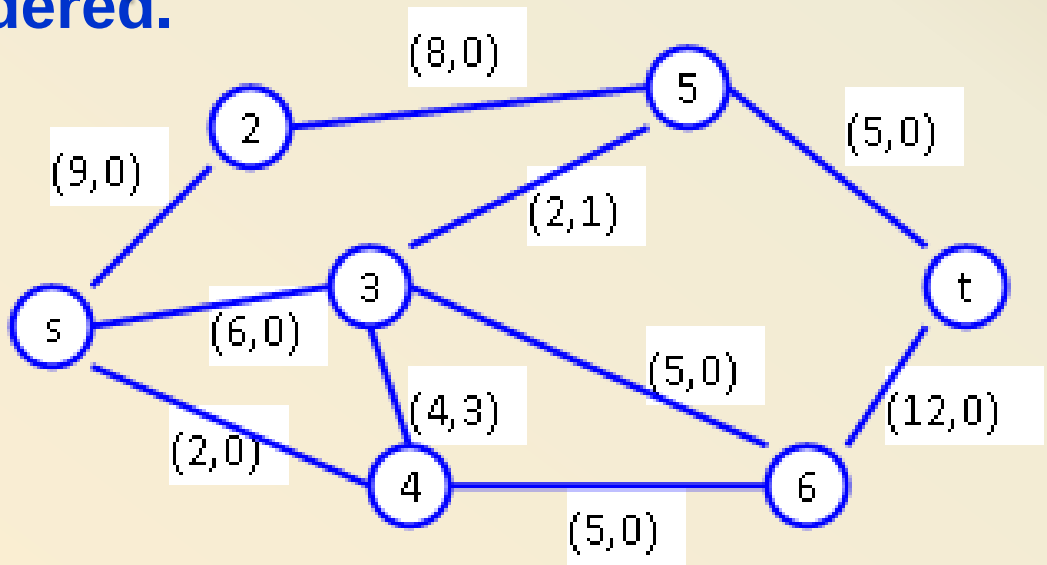


Maximum Flow Minimum Cut Theorem

- **This theorem states that in a flow network, the maximum value of flow from source s to sink t is equal to the minimum value of the capacities of all the cuts in the network that separates source from the sink.**

- **Note: if capacity of cut is C(p, ṕ) only capacities from s to t should be considered.**

- **Question: Find the maximum possible flow in this flow network.**



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Maximal Flow Problems (Continued)

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Maximum Flow Minimum Cut Theorem

- **This theorem states that in a flow network, the maximum value of flow from source s to sink t is equal to the minimum value of the capacities of all the cuts in the network that separates source from the sink.**

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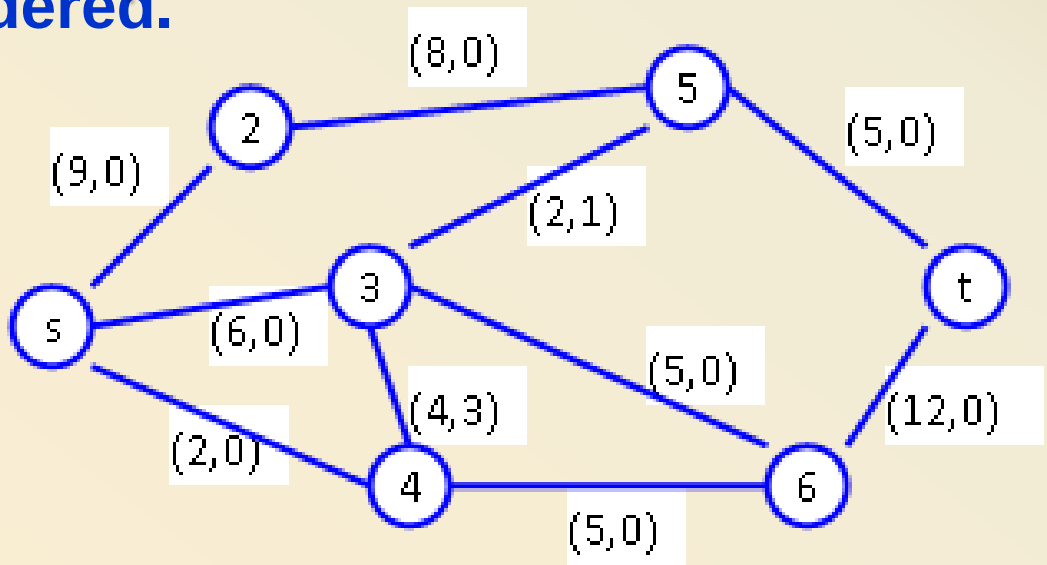
- **Question: Find the maximum possible flow in this flow network.**



Network Flow Example

Consider the graph given below (in terms of edge, forward flow capacity, and backward flow capacity). Assume lossless flow and infinite node capacity for each node.

- ***s-2(9,0); s-3(6,0); s-4(2,0); 2-5(6,0); 3-4(4,3);***
- ***3-5(2,3); 3-6(5,0); 4-6(5,0); 5-t(5,0); 6-t(12,0)***

- ***Draw the network flow diagram and***
- ***find the maximal flow from node s to node t using a Network flow algorithm.***

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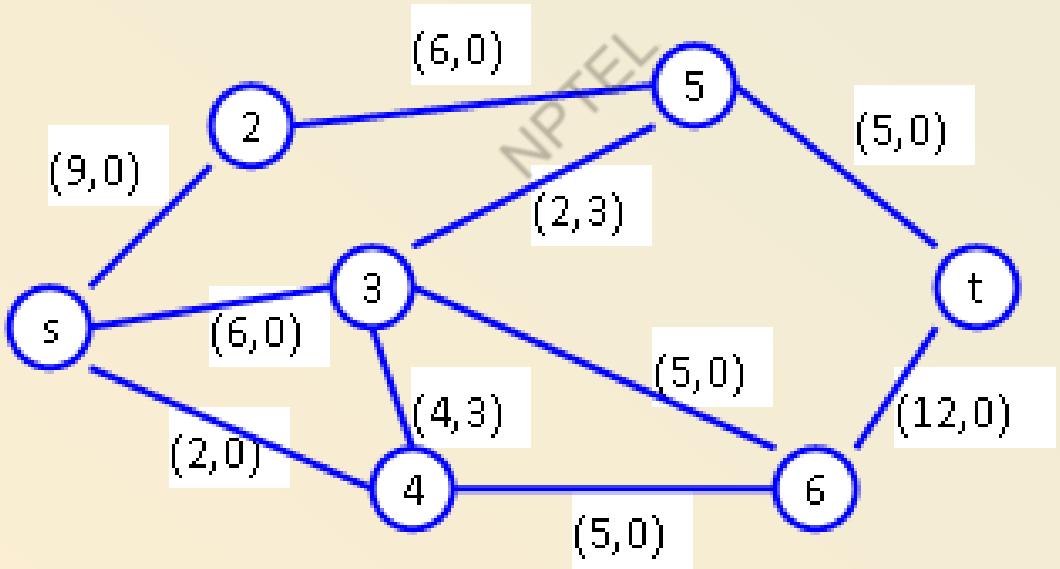
Network Flow Diagram

- ***s-2(9,0); s-3(6,0); s-4(2,0); 2-5(6,0); 3-4(4,3);***
- ***3-5(2,3); 3-6(5,0); 4-6(5,0); 5-t(5,0); 6-t(12,0)***

- ***Draw the network flow diagram***



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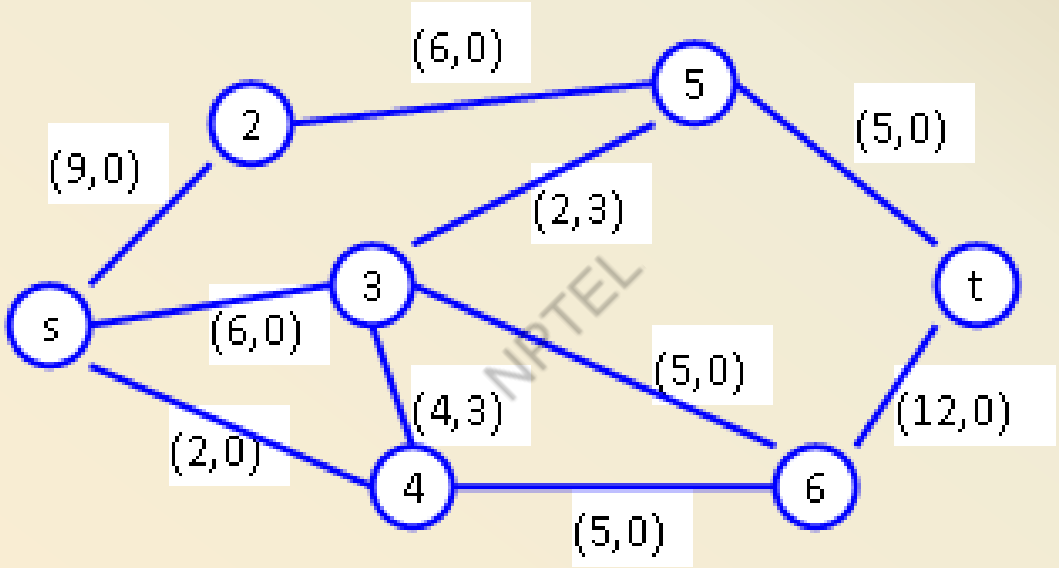
Flow Augmenting Paths

- ***Find the maximal flow from node s to node t using a Network flow algorithm.***

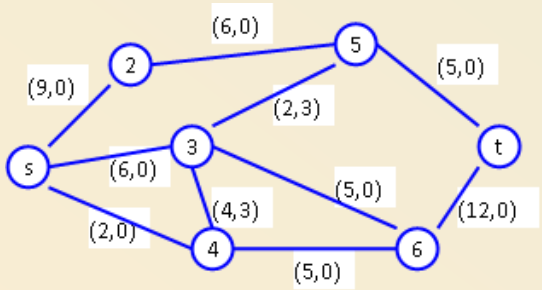


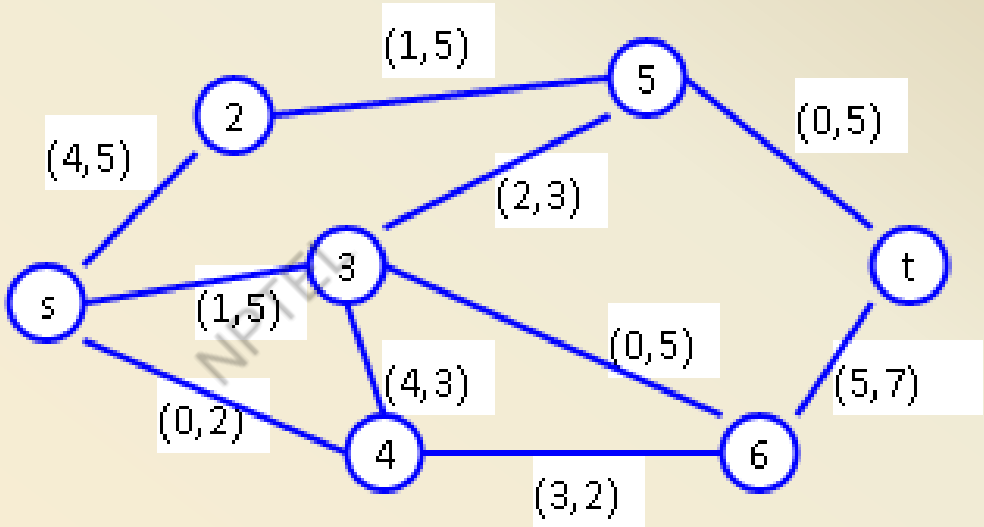
Flow-augmenting paths:

- **FAP1 = s-2-5-t; Maximum Flow possible is 5.**
- **FAP2 = s-3-6-t, Maximum Flow possible is 5.**
- **FAP3 = s-4-6-t, Maximum flow Possible is 2.**

Revising Flow Capacities

- ***Find the maximal flow from node s to node t using a Network flow algorithm.***



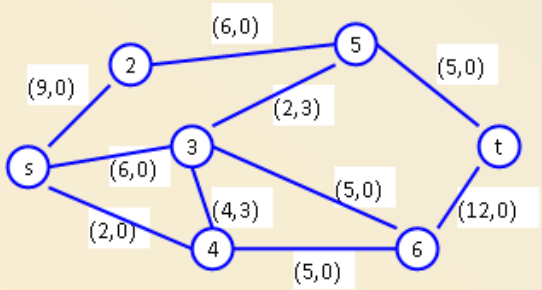


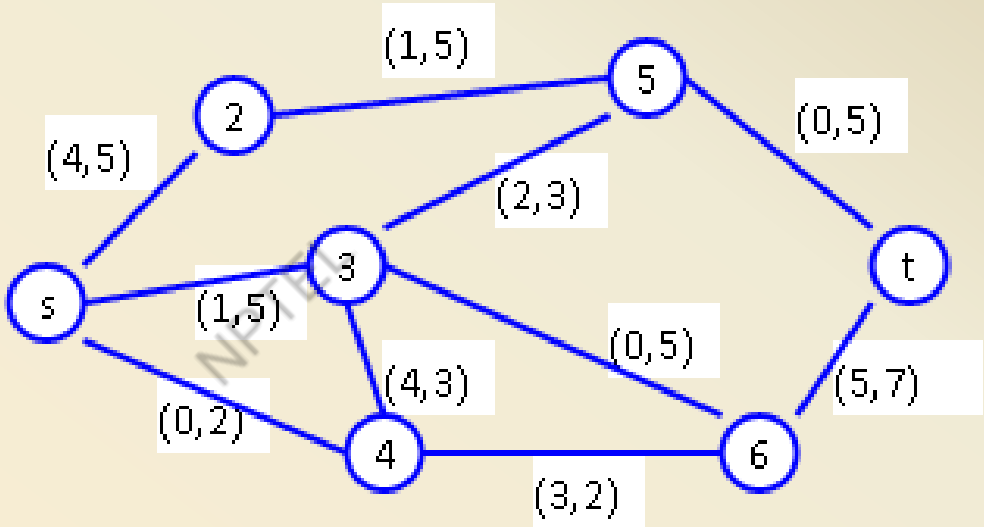
Revised Capacities after flow is assigned as per the three flow-augmenting paths.

- **FAP1 = s-2-5-t; Maximum Flow possible is 5.**
- **FAP2 = s-3-6-t, Maximum Flow possible is 5.**
- **FAP3 = s-4-6-t, Maximum flow Possible is 2.**

New Flow Augmenting Paths

- ***Find the maximal flow from node s to node t using a Network flow algorithm.***



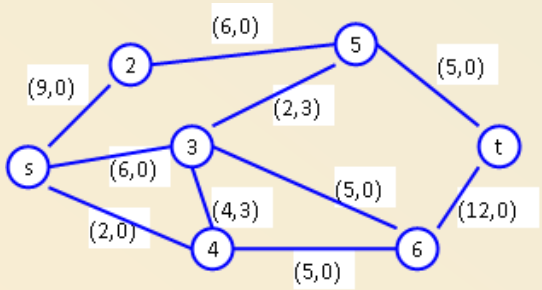


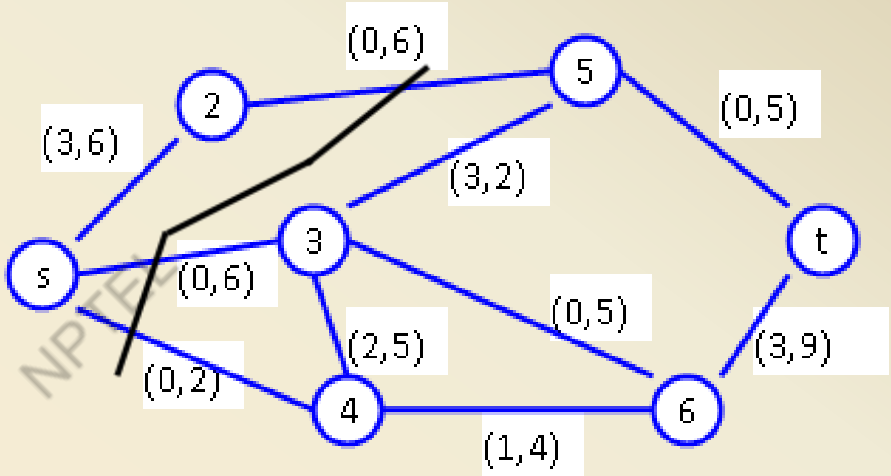
New Flow augmenting paths:

- **FAP4 = s-2-5-3-4-6-t; Flow = 1.**
- **Also, FAP5 = s-3-4-6-t; Flow = 1**

Final Revised Capacities

- ***Find the maximal flow from node s to node t using a Network flow algorithm.***





Flow Capacities after assigning Flow augmenting paths:

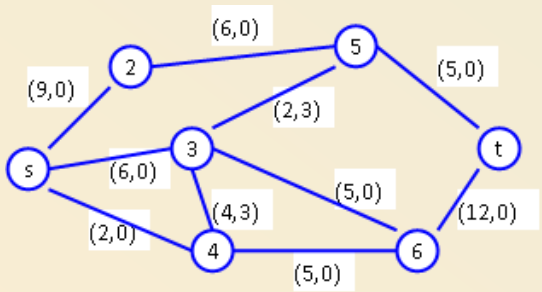
- **FAP4 = s-2-5-3-4-6-t; Flow = 1.**
- **Also, FAP5 = s-3-4-6-t; Flow = 1**

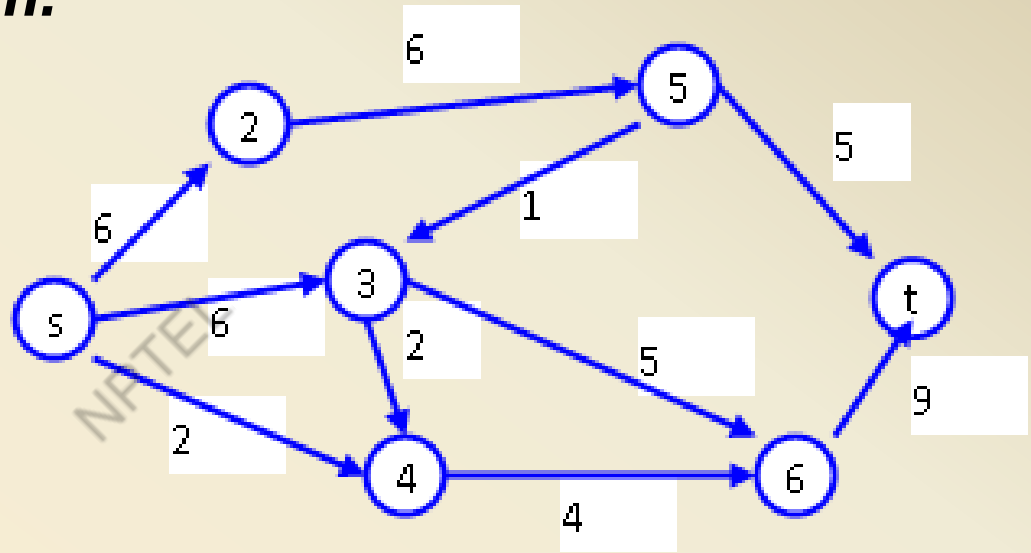
No more flow is possible!

Check this from the cutset shown.

Final Flow Pattern

- ***Find the maximal flow from node s to node t using a Network flow algorithm.***

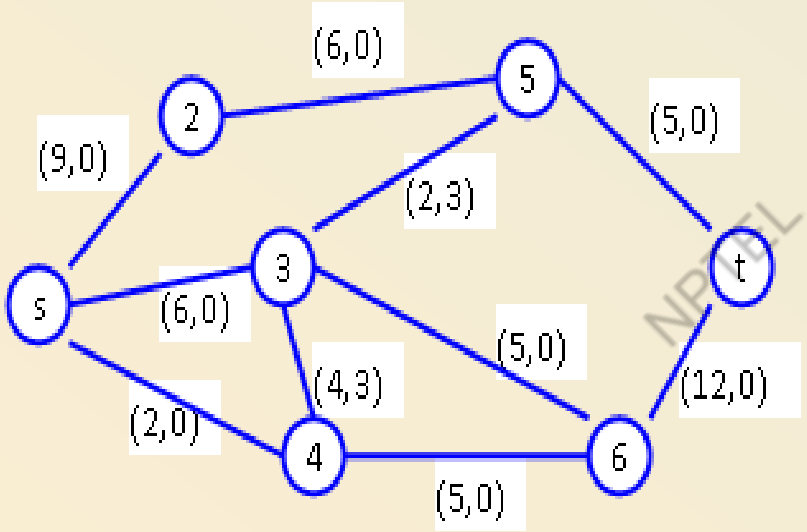




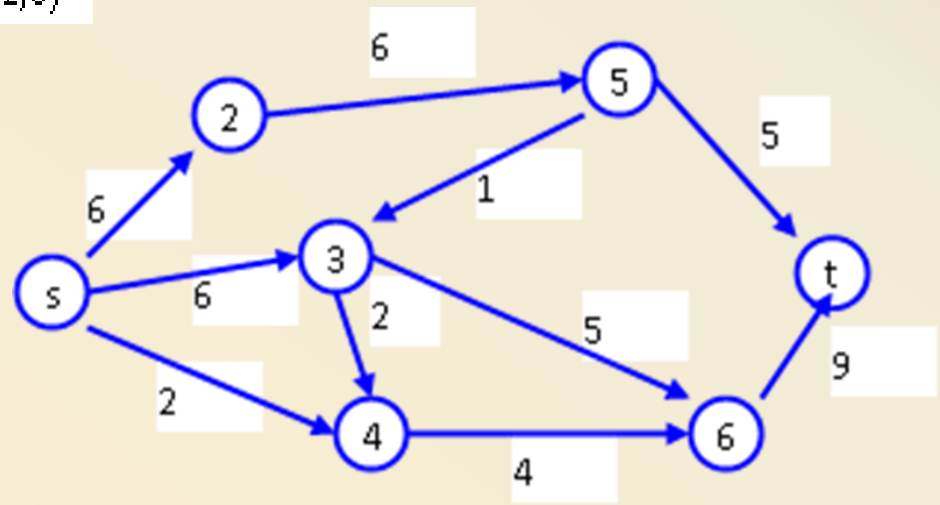
- **The final maximal flow pattern is shown in the diagram.**
- **Maximal flow = 14 at s or t**

Limited Node Capacity

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**

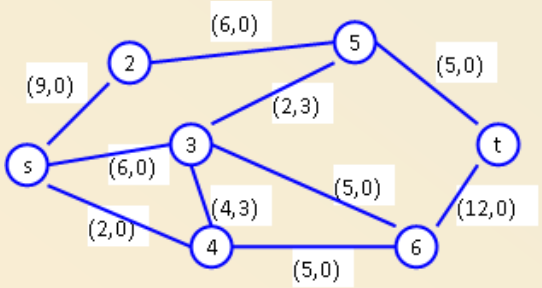


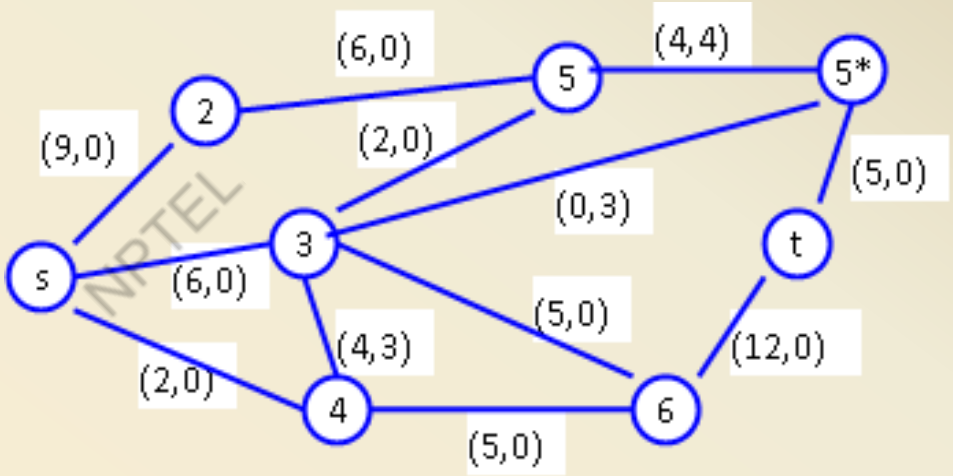
- ***The final maximal flow pattern is shown in the diagram.***
- ***Maximal flow = 14 at s or t***



Limited Node Capacity

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**

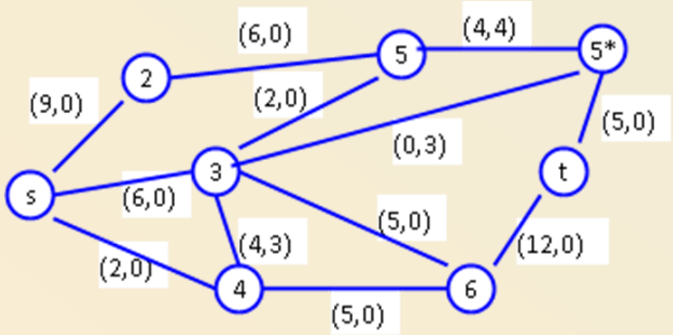


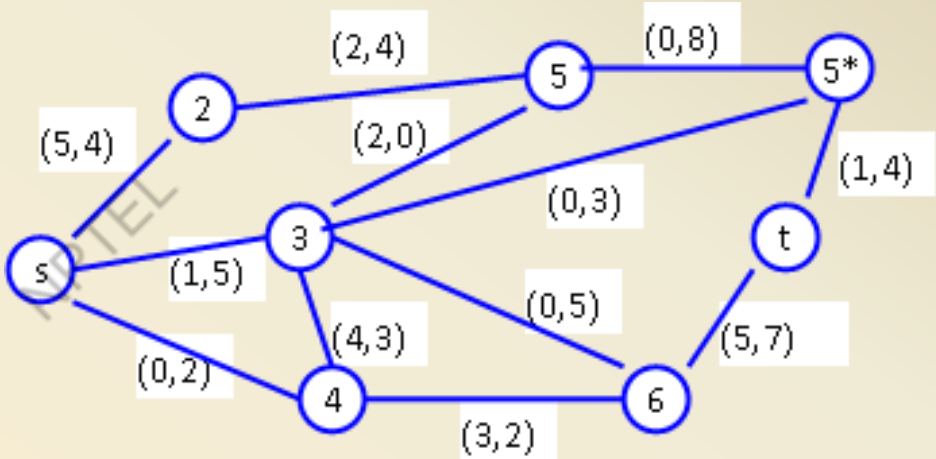


- **Look at the new network diagram. Observe the changes.**

Revising Flow Capacities

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**

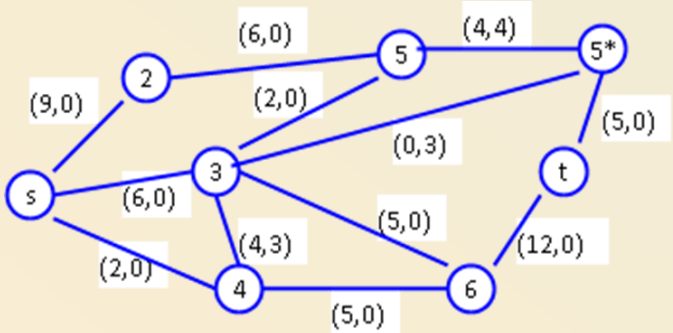


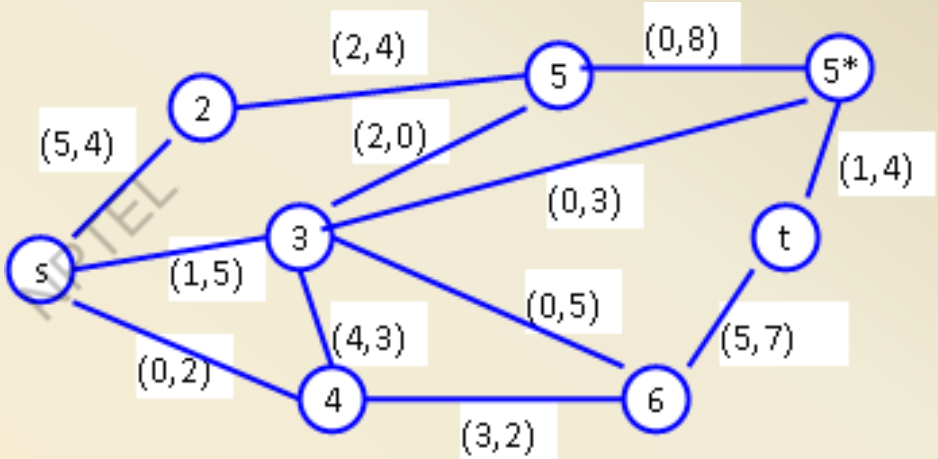


- **FAP1 = s-2-5-5*-t;** **Maximum Flow = 4;**
- **FAP2 = s-3-6-t;** **Maximum Flow = 5;**
- **FAP3= s-4-6-t;** **Maximum Flow = 2;**

New Flow Augmenting Paths

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**

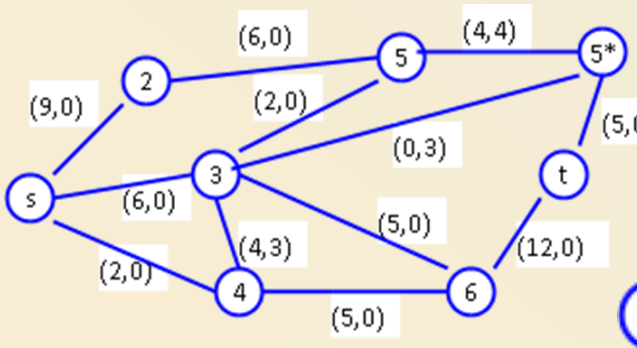


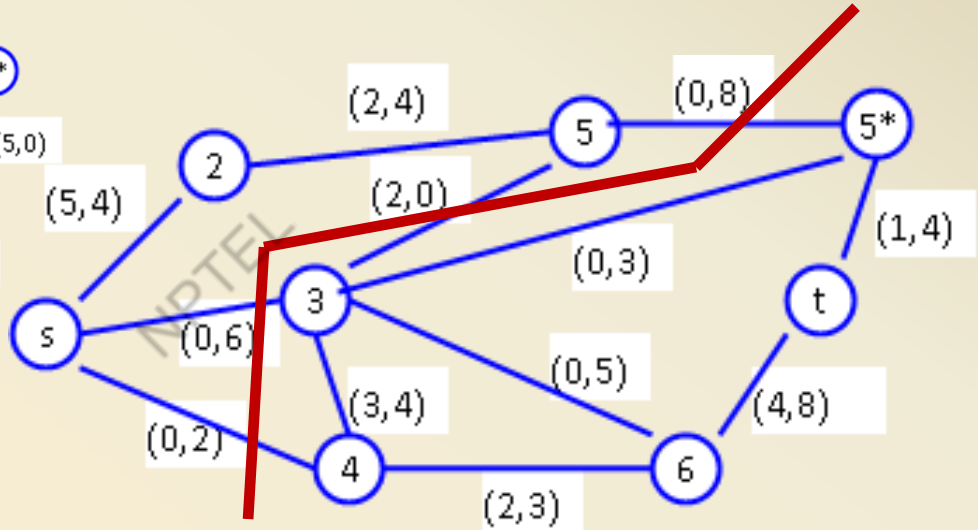


- **Next, we may have only one more flow-augmenting path.**
- **FAP4 = s-3-4-6-t; Flow = 1**

Final Revised Capacities

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**

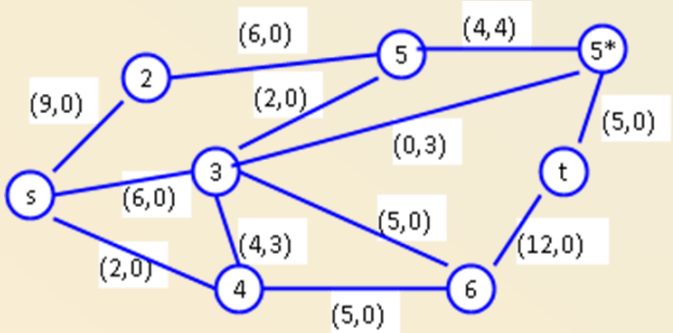


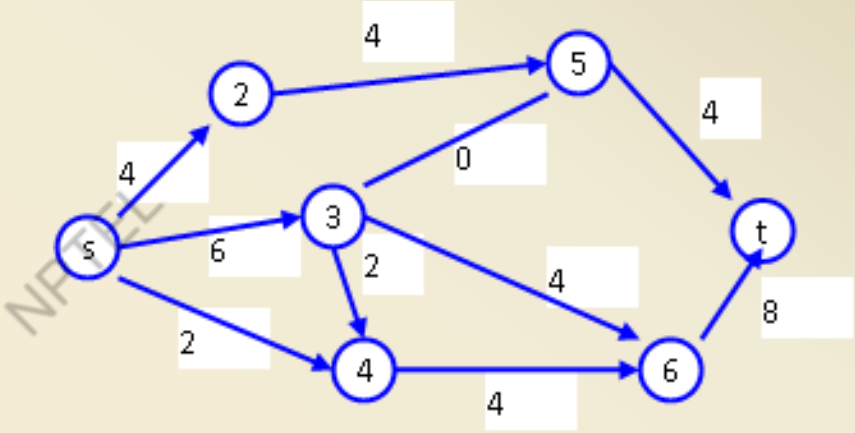


- **No more flow is possible!**
- **Check this from the cutset shown.**
- **Note, the cutset requires flow from 5 to 3 in 3-5 link, which is not possible!**

Final Flow Pattern

- **Now consider that Node 5 has a limited flow capacity of 4 units only. Find out the maximal flow from Node s to Node t using this additional constraint.**





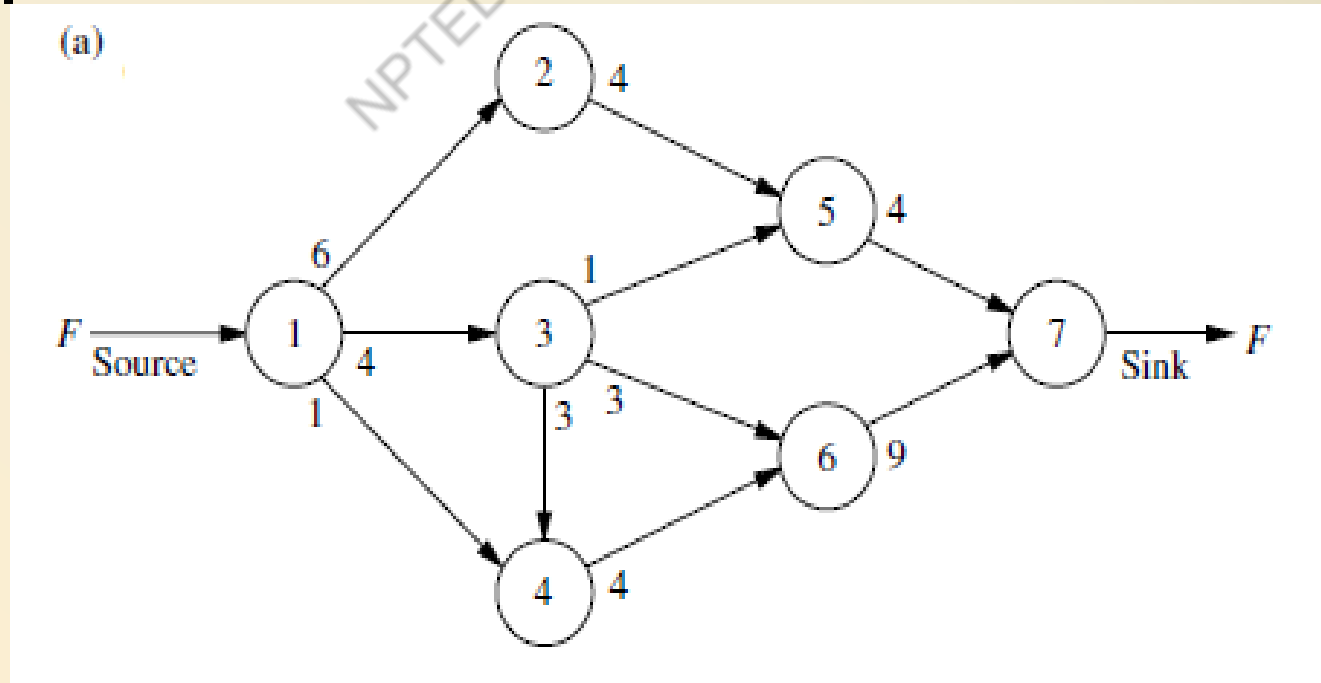
- **The final maximal flow pattern is shown above**
- **Maximal flow = 12 at s or t**

Another Maximal Flow Problem

Consider the graph in terms of forward flow capacity of each arc in the direction shown. Assume infinite node capacities; and zero reverse flow capacity and lossless flow for each arc.

Find all the cuts and their capacities. Use 'Maximum Flow Minimum Cut Theorem' to determine the maximal flow from node 1 to node 7.

Find the flow pattern in the network with respect to maximal flow from the node 1 to node 7.



Module 39

Shortest Path Problems

NPTEL

Shortest Path Problems

Shortest Path problems abound in literature. Some examples are:

1. **Shortest paths in road networks**
2. **Shortest paths in communication networks**
3. **Shortest paths in graphs – a graph can represent different problems. The shortest path may have different meaning in different graphs.**

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Some Shortest Path Algorithms

- **Dijkstra's algorithm:** single-source shortest path problems.
- **Bellman–Ford algorithm:** single-source problem with negative edge weights or distances.
- **A* search algorithm:** single pair shortest path using heuristics to speed up the search.
- **Floyd's or Floyd–Warshall algorithm:** all pairs of shortest paths.

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Dijkstra's Algorithm

Dijkstra's algorithm is applicable for single-source shortest path problems.

- This algorithm is based on the well-known concept of Dynamic Programming.
- Dynamic programming is used to solve a wide variety of discrete optimization problems such as scheduling, string-editing, packaging, and inventory management.
- Break problems into sub-problems and combine their solutions into solutions to larger problems.
- In contrast to divide-and-conquer, there may be relationships across sub-problems.

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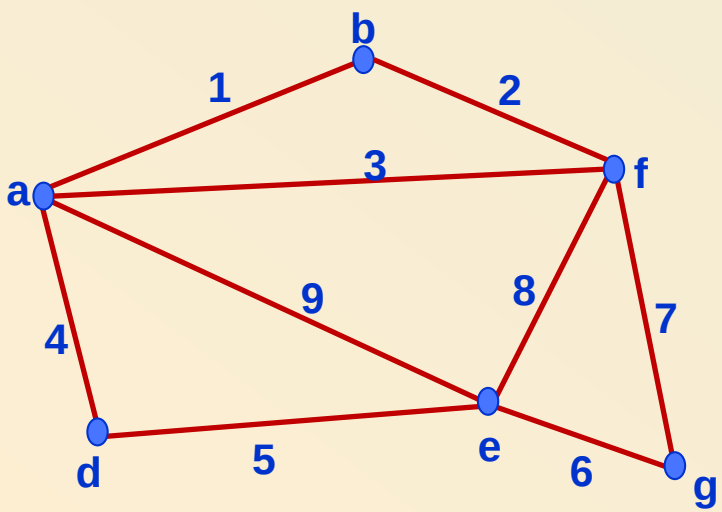
Bellman's Principle of Optimality

Richard Bellman's Principle of Optimality:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.



- **Suppose, we started from 'a' for reaching 'g'**
- **Presently, we are at 'f'**
- **Then, no matter how we reached 'f' from 'a' – we should optimally reach from 'f' to 'g' – How to do this?**
- **Go directly! – Distance is 7.**

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Stagecoach Problem

- A fortune seeker in Missouri travels to California by stagecoach to join the gold rush through unsettled country of serious danger.
- Although his starting point and destination were fixed, he had a lot of choice in his route of travel.
- Being prudent to consider safety, he found out that the safest route should be the one with the cheapest total life insurance policy.
- It is because the cost of the policy must have been worked out after a carefully evaluation of the safety of every run of the stagecoach travel.

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Stagecoach Problem

	B	C	D
A	2	4	3

	E	F	G
B	7	4	6
C	3	2	4
D	4	1	5

	H	I
E	1	4
F	6	3
G	3	3

	J
H	3
I	4

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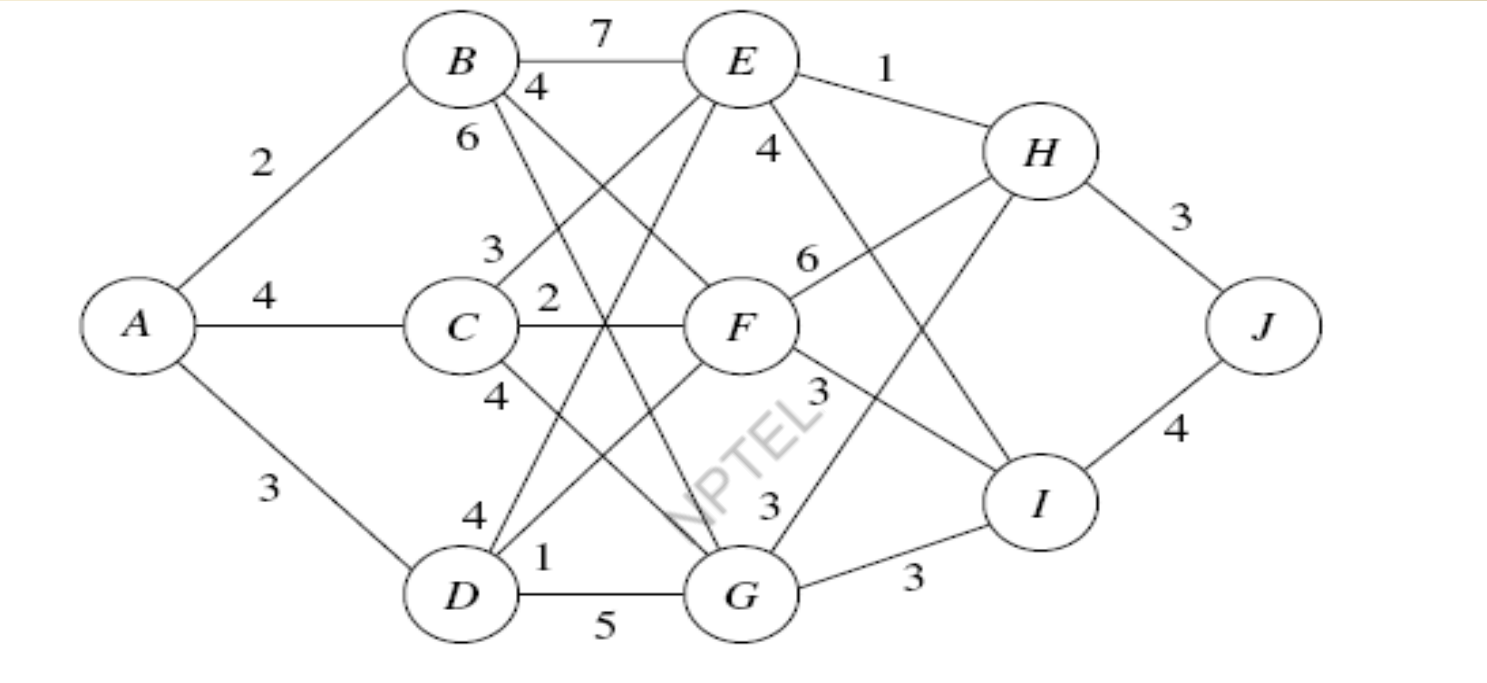
A: Starting city; **J: Destination city**

B,C,D: Stage 1 cities; **E,F,G: Stage 2 cities;**

H,I: Stage 3 cities

The cost of policies are indicated. One should find the shortest path.

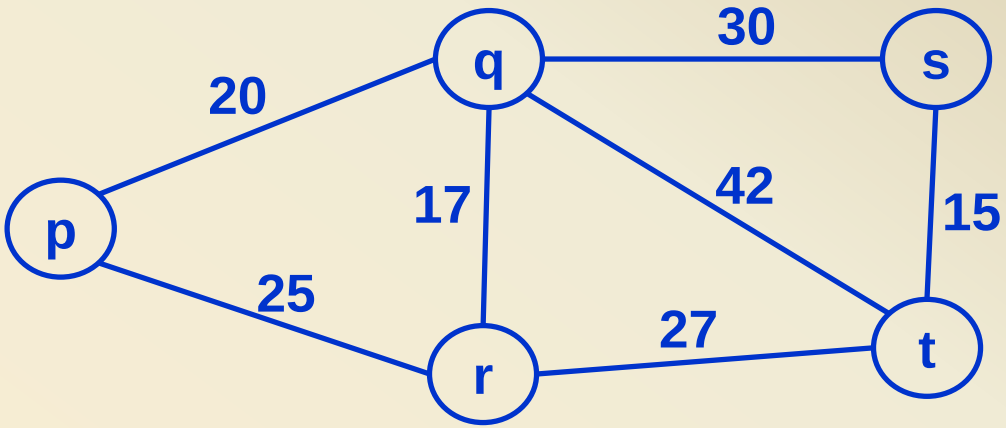
Stagecoach Problem



How do we use Bellman's Principle here?

Will the shortest path of the network be the optimal solution?

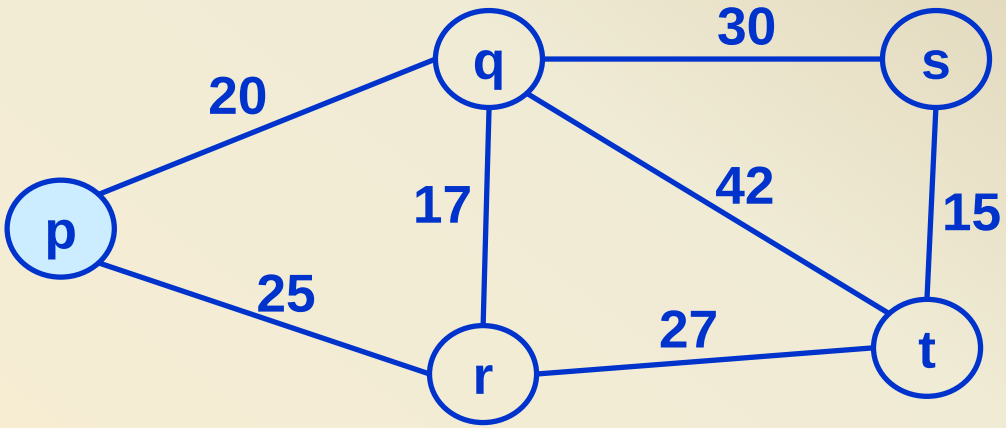
A Shortest Path Problem



Node	p	q	r	s	t
Value	0	∞	∞	∞	∞

NPTEL

A Shortest Path Problem



Node	p	q	r	s	t
Value	0	∞	∞	∞	∞

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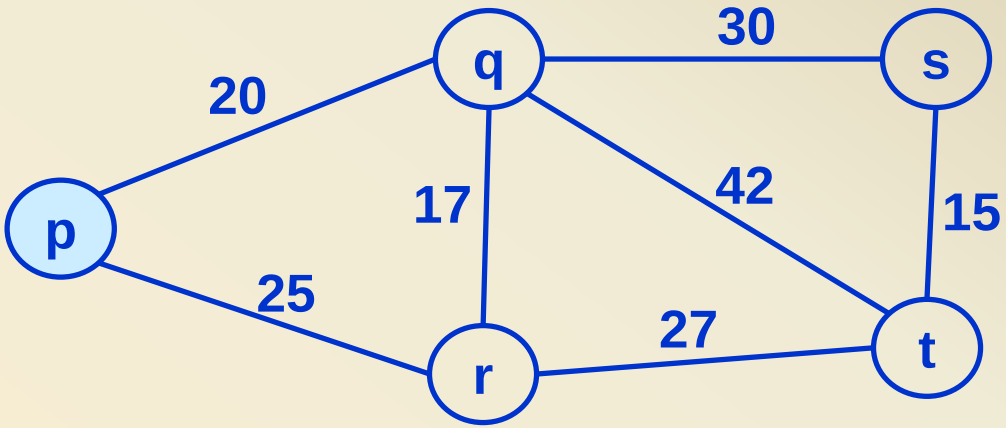
Make p permanent

List of permanent nodes: {p}

A Shortest Path Problem



Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞

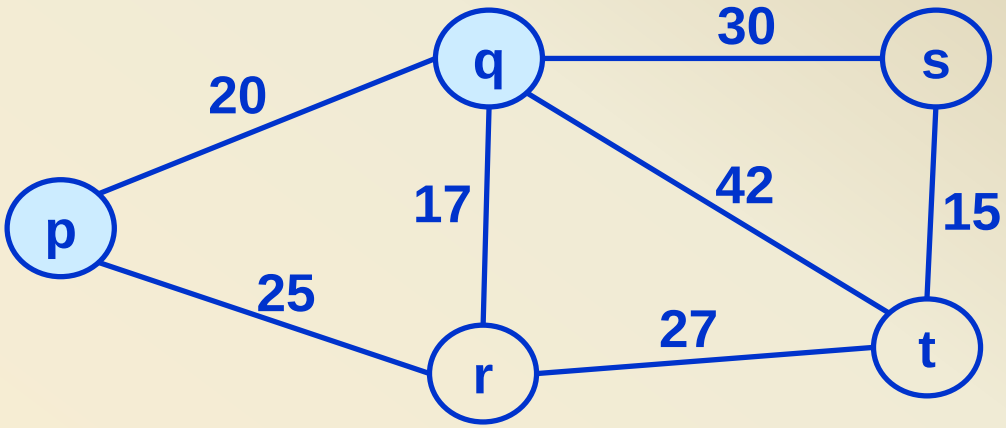
List of permanent nodes: **{p}** ***p: last permanent node***

Note: Value for Node 'q'= Min [∞, (0+20)] = 20

Value for Node 'r'= Min [∞, (0+25)] = 25

NPTEL

A Shortest Path Problem



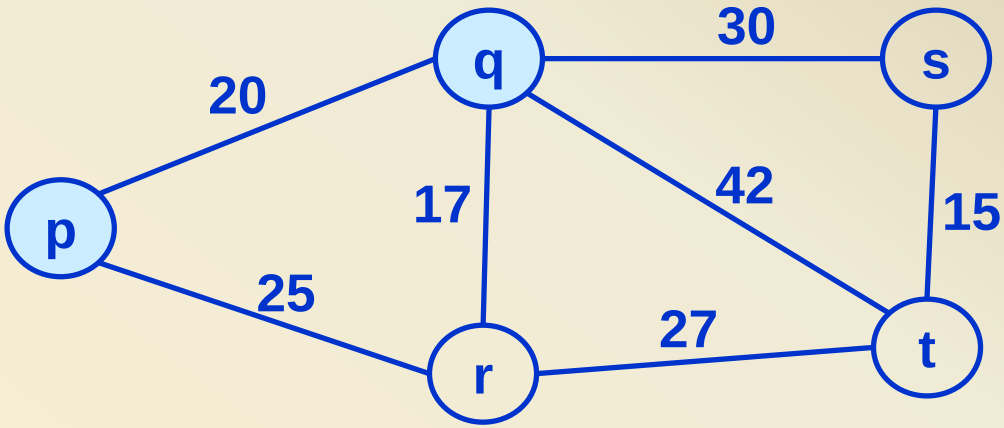
Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞

Make q permanent

List of permanent nodes: {p, q}

NPTEL

A Shortest Path Problem



Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62

NPTEL

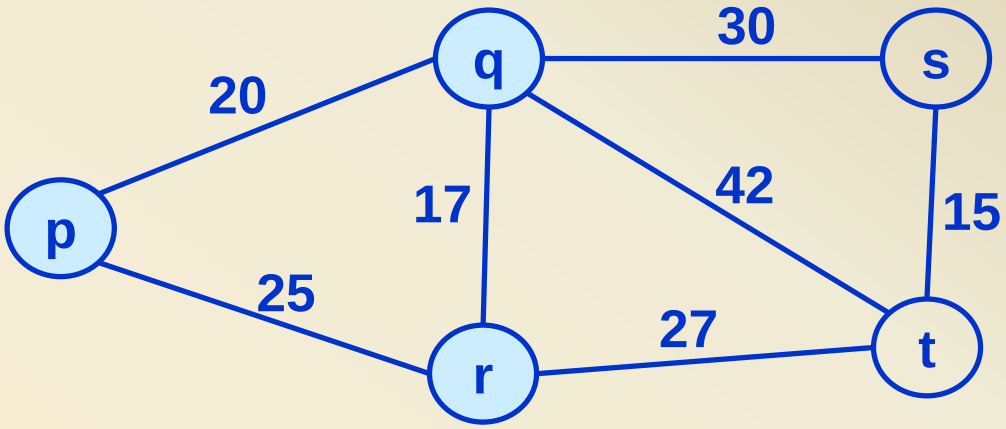
List of permanent nodes: {p, q} *q: last permanent node*

Note: Value for Node 'r'= Min [25, (20+17)] = 25

Value for Node 's'= Min [∞, (20+30)] = 50

Value for Node 't'= Min [∞, (20+42)] = 62

A Shortest Path Problem



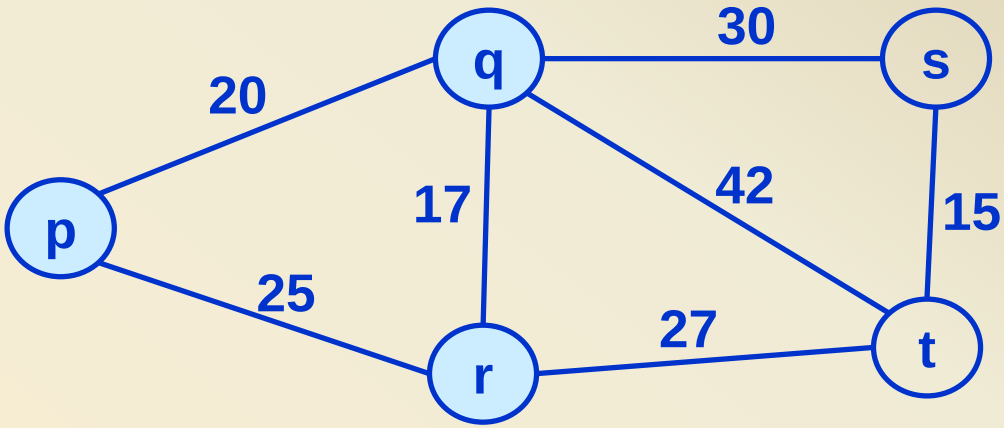
Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62

NPTEL

Make r permanent

List of permanent nodes: {p, q, r}

A Shortest Path Problem



Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62
				50	52

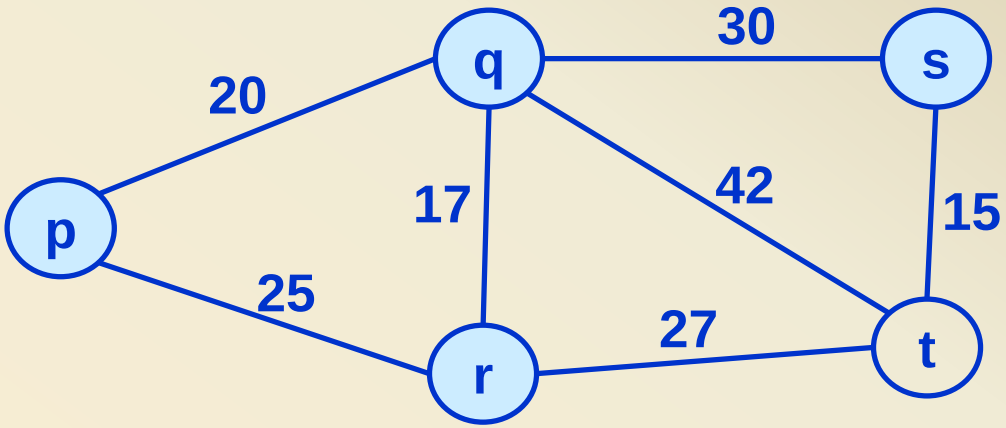
NPTEL

List of permanent nodes: {p, q, r} r: *last permanent node*

Note: Value for Node 's'= Min [50, (25+ ∞)] = 50

Value for Node 't'= Min [∞, (25+27)] = 52

A Shortest Path Problem



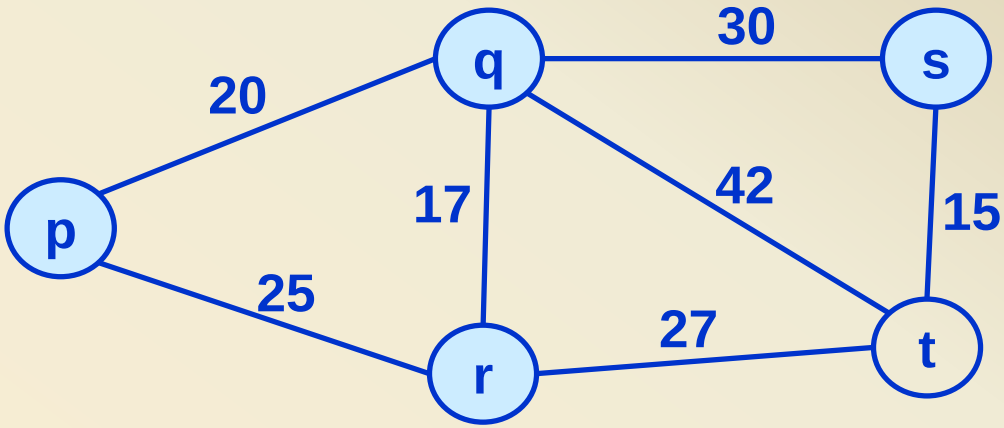
Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62
				50	52

NPTEL

Make s permanent

List of permanent nodes: {p, q, r, s}

A Shortest Path Problem



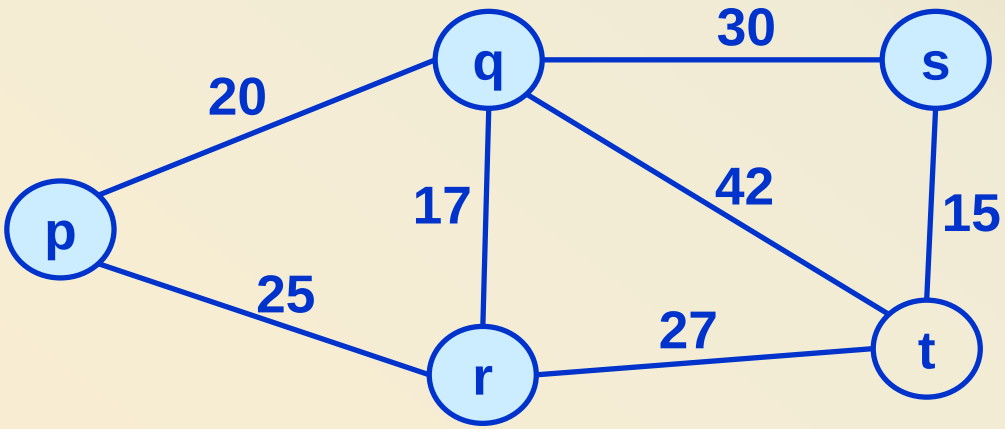
Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62
				50	52
					52

NPTEL

List of permanent nodes: {p, q, r, s} *s: last permanent node*

***Note**: Value for Node 't'= Min [52, (50+15)] = 52*

A Shortest Path Problem



Shortest Path from r to t

't' is made permanent from 'r'
'r' is made permanent from 'p'

So, Shortest Path is p-r-t, length 52

Node	p	q	r	s	t
Value	0	∞	∞	∞	∞
		20	25	∞	∞
			25	50	62
				50	52
					52

NPTEL

Make t permanent – All nodes are permanent! Stop.

List of permanent nodes: {p, q, r, s, t}

Shortest Path Problems (Continued)

NPTEL

Dijkstra's Algorithm

Step 1

Assign label 0 to starting node s and all other nodes a label ∞. Make s a permanent node. Let p = s; p is the last node made permanent.

Step 2

Let d_{pk} be the distance from node p to all other non-permanent nodes k. Recompute l_k, the label of each node k as:

$$l_k = \text{Min}\{l_k,(l_p+d_{pk})\}$$

Step 3

If the ending node t is made permanent, stop. Lt is the shortest path from s to t.

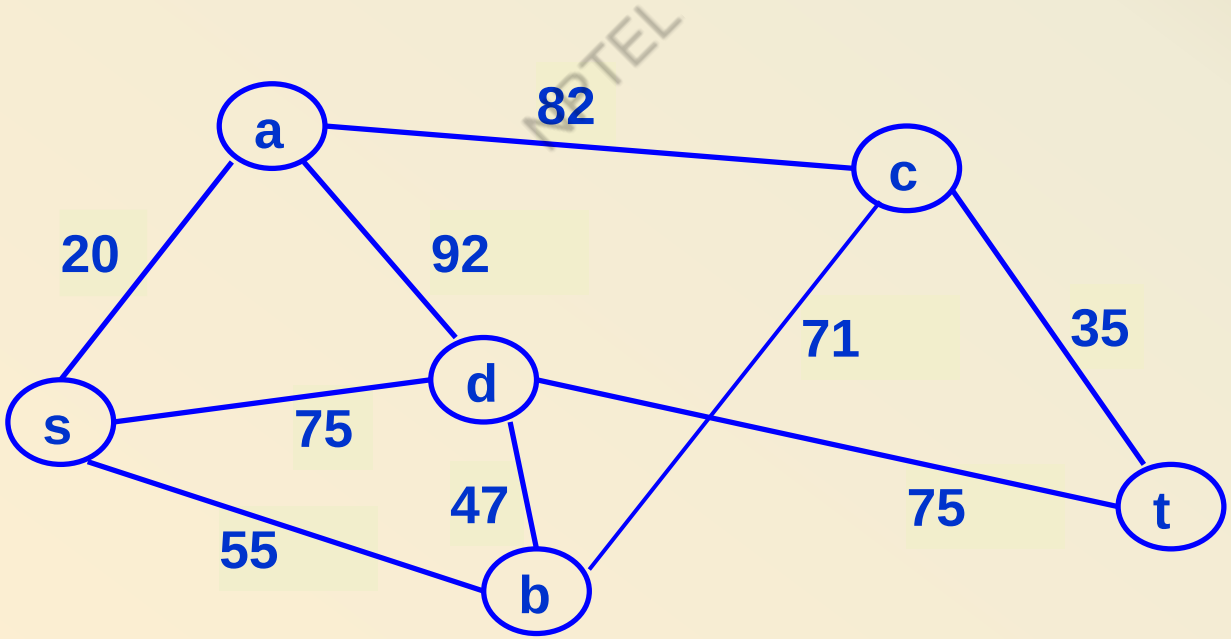
Else go to step 2.

NPTEL

Dijkstra's Algorithm

Consider a network with distances:

- **s-a : 20; s-b: 55; s-d : 75; a-c : 82; a-d : 92; b-c: 71; b-d: 47; c-t : 35; d-t : 75.**
- **Find out the shortest path between s and t.**

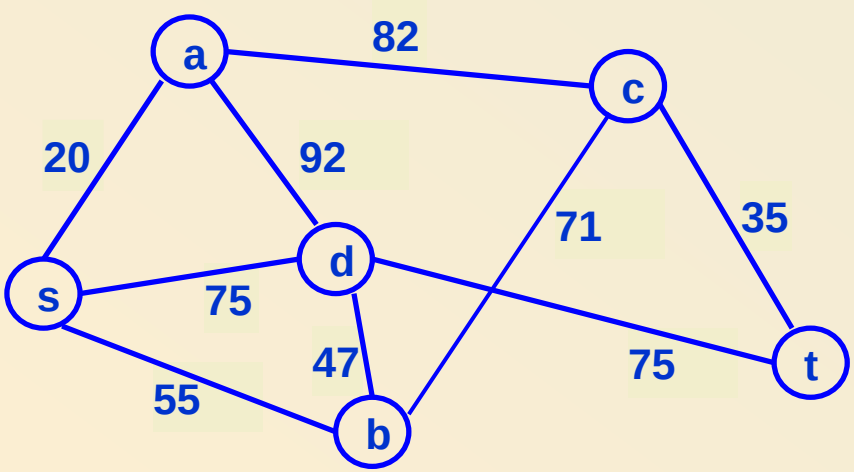


NPTEL

Dijkstra's Algorithm

Distance Matrix	s	a	b	c	d	t
s	-	20	55	-	75	-
a	20	-	-	82	92	-
b	55	-	-	71	47	-
c	-	82	71	-	-	35
d	75	92	47	-	-	75
t	-	-	-	35	75	-

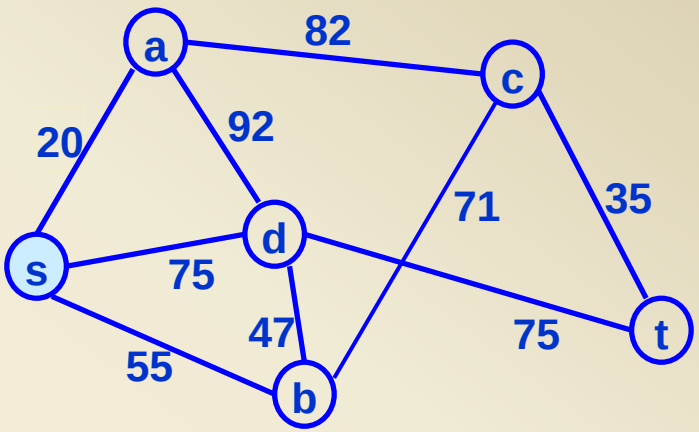
NPTEL



Dijkstra's Algorithm

Step 0	s	a	b	c	d	t
Temp label l(i)	0	∞	∞	∞	∞	∞
Permanent?	Yes	No	No	No	No	No
Made from	-	-	-	-	-	-

NPTEL



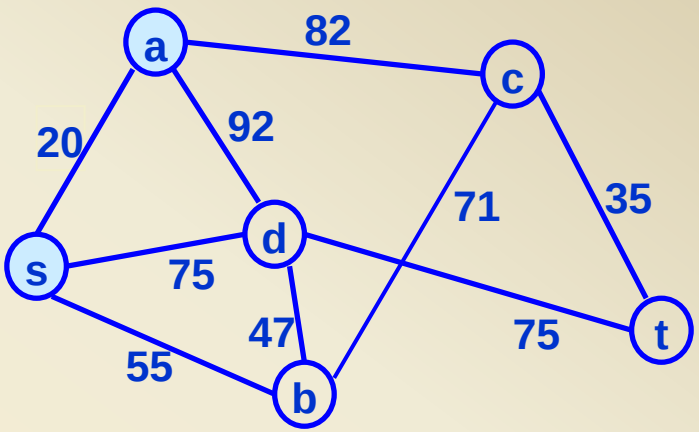
Initially, all the nodes are assigned an ∞ label except the starting node s which is assigned a label of 0.

's' is made a permanent node.

Dijkstra's Algorithm

Step 1	s	a	b	c	d	t
Temp label l(i)	0	20	55	∞	75	∞
Perma nent?	Yes	Yes	No	No	No	No
Made from	-	s	s	-	s	-

NPTEL



l(a) = Min {l(a), l(s) + d(s,a)} = Min {∞, 0+20} = 20 **

l(b) = Min {l(b), l(s) + d(s,b)} = Min {∞, 0+55} = 55

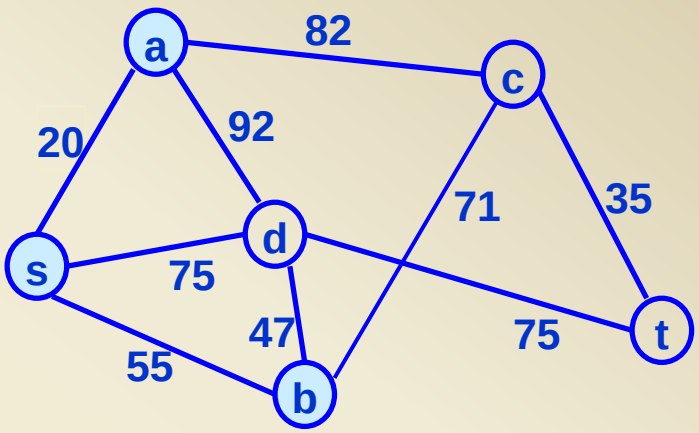
l(d) = Min {l(d), l(s) + d(s,d)} = Min {∞, 0+75} = 75

** 'a' is now made permanent from node 's'

Dijkstra's Algorithm

Step 2	s	a	b	c	d	t
Temp label l(i)	0	20	55 ↗	102	75	∞
Perma nent?	Yes	Yes	Yes	No	No	No
Made from	-	s	s	a	s	-

NPTEL



l(b) = Min {l(b), l(a) + d(a,b)} = Min {55, 20+∞} = 55 **

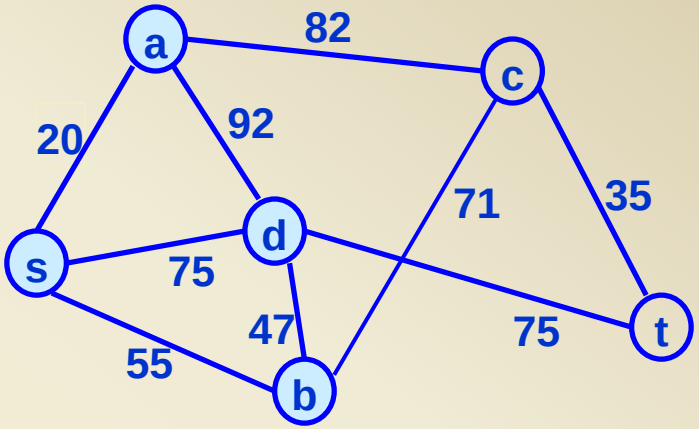
l(c) = Min {l(c), l(a) + d(a,c)} = Min {∞, 20+82} = 102

l(d) = Min {l(d), l(a) + d(a,d)} = Min {75, 20+92} = 75

** 'b' is now made permanent from node 's'

Dijkstra's Algorithm

Step 3	s	a	b	c	d	t
Temp label l(i)	0	20	55	102	75	∞
Perma nent?	Yes	Yes	Yes	No	Yes	No
Made from	-	s	s	a	s	-



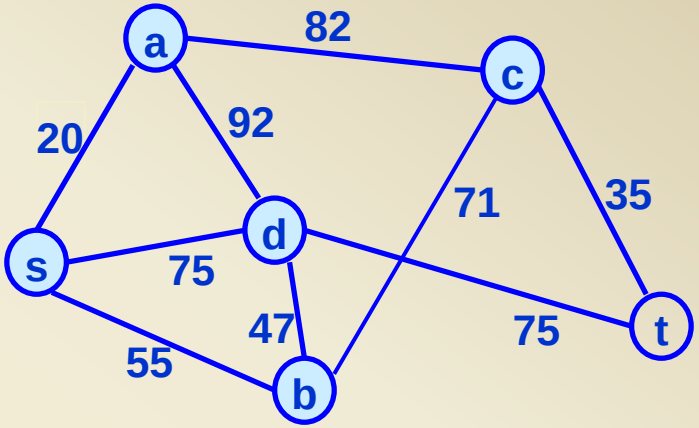
NPTEL

l(c) = Min {l(c), l(b) + d(b,c)} = Min {102, 55+71} = 102
l(d) = Min {l(d), l(b) + d(b,d)} = Min {75, 55+47} = 75 **

** 'd' is now made permanent from node 's'

Dijkstra's Algorithm

Step 4	s	a	b	c	d	t
Temp label l(i)	0	20	55	102	75	150
Permanent?	Yes	Yes	Yes	Yes	Yes	No
Made from	-	s	s	a	s	-



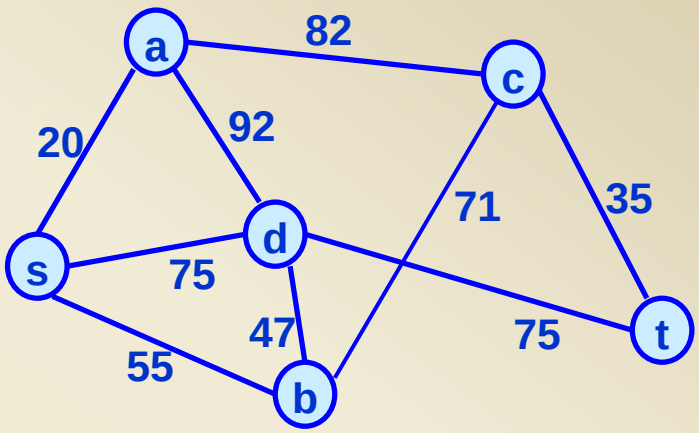
NPTEL

l(c) = Min {l(c), l(d) + d(d,c)} = Min {102, 75+∞} = 102 **
l(t) = Min {l(t), l(d) + d(d,t)} = Min {∞, 75+75} = 150

** 'c' is now made permanent from node 'a'

Dijkstra's Algorithm

Step 5	s	a	b	c	d	t
Temp label l(i)	0	20	55	102	75	137
Perma nent?	Yes	Yes	Yes	Yes	Yes	Yes
Made from	-	s	s	a	s	c



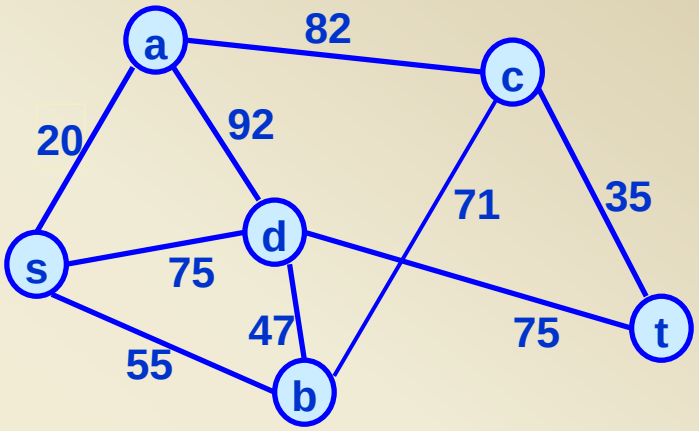
NPTEL

l(t) = Min {l(t), l(c) + d(c,t)} = Min {150, 102+35} = 137**

** 't' is now made permanent from node 'c'

Dijkstra's Algorithm

Step 6	s	a	b	c	d	t
Temp label l(i)	0	20	55	102	75	137
Permanent?	Yes	Yes	Yes	Yes	Yes	Yes
Made from	-	s	s	a	s	c



NPTEL

Obtaining Shortest Path

't' is made permanent from 'c'. 'c' is made permanent from 'a', 'a' is made permanent from 's'.

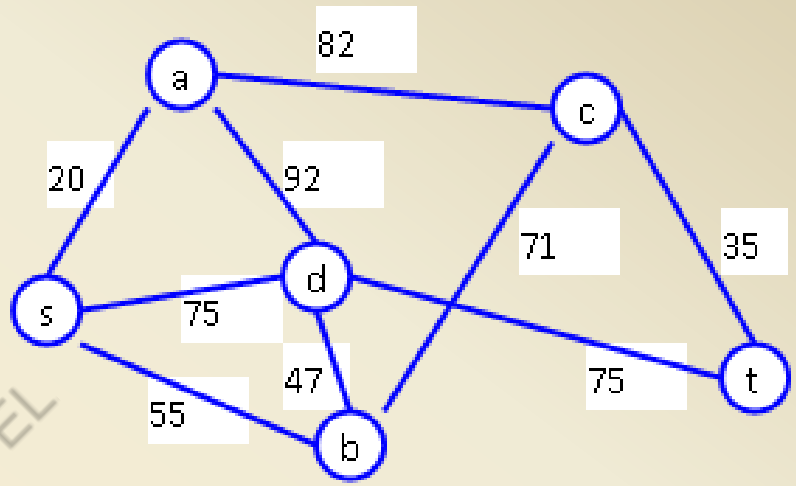
Hence, shortest path from 's' to 't' is 's-a-c-t' and the shortest distance is 137.

Shortest Path Exercise

Find Shortest Path in the graph shown from node s to node t using:

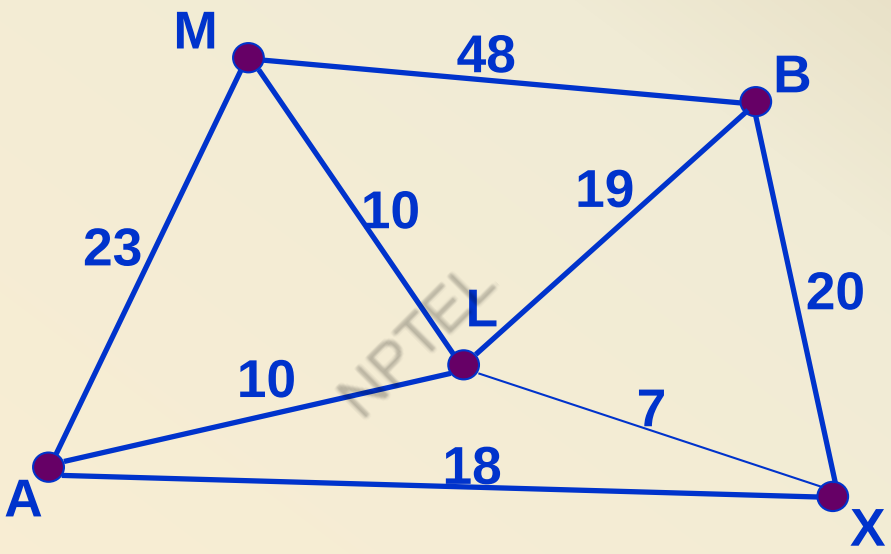
- ***Dijkstra's Algorithm***

Also, find the length of the shortest path.



NPTEL

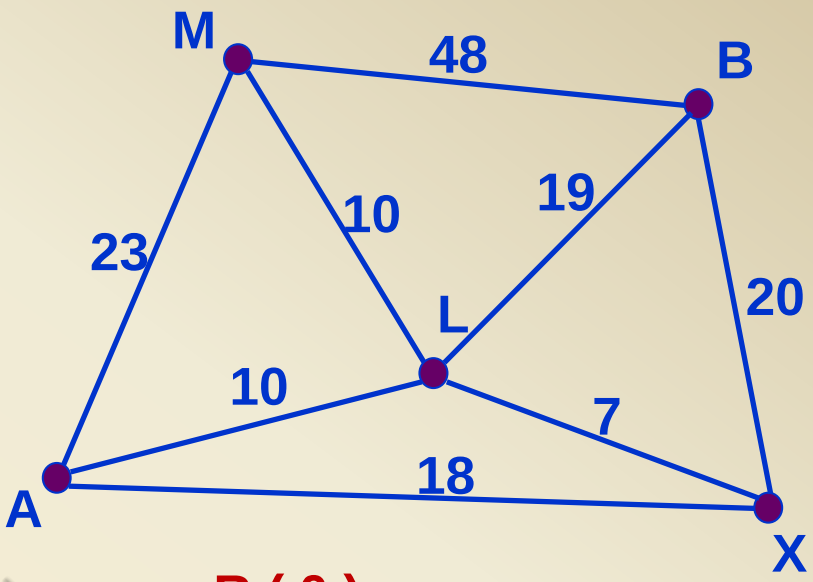
Floyd's Algorithm – Shortest Path among All Pair of Nodes



Floyd's Algorithm is another algorithm to solve shortest path problems. This algorithm gives shortest path from any node to any other node by using a matrix method.

73

Floyd's Algorithm



D (0)

	A	M	L	B	X
A	∞	23	10	∞	18
M	23	∞	10	48	∞
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

Distance Matrix

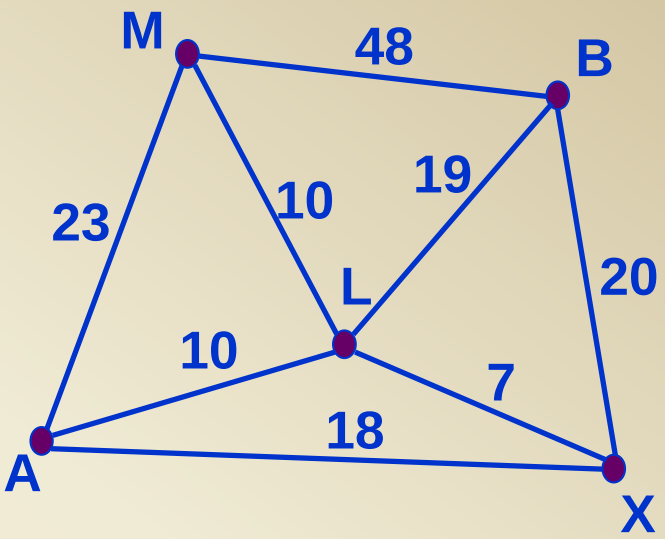
R (0)

	A	M	L	B	X
A	1	2	3	4	5
M	1	2	3	4	5
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

Route Matrix

NPTEL

Floyd's Algorithm



We highlight the 1st column and row of D(0) and 1st column of R(0) and compare all other items with the sum of the items highlighted in the same row and column. If the sum is less than the item then it should be replaced with the sum.

NPTEL

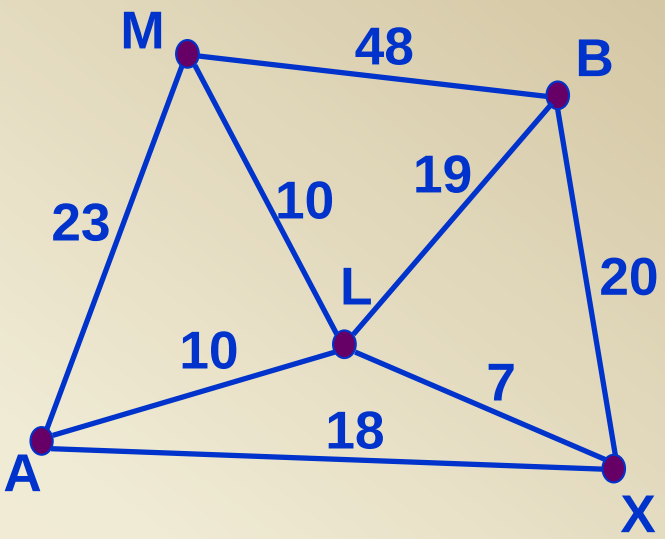
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	∞	10	48	∞
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	2	3	4	5
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



See the highlighted item in red – the sum of yellow highlighted items in corresponding row and column is 23 + 23 = 46 - less than the original value infinity. So the highlighted item in red will be replaced by 46.

NPTEL

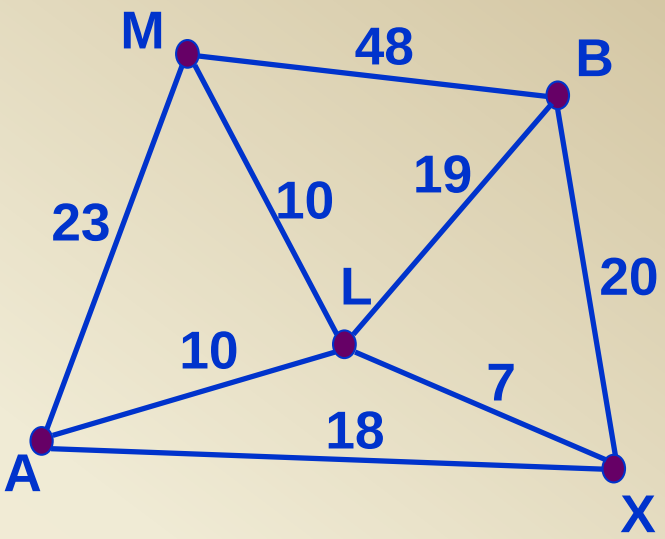
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	∞	10	48	∞
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	2	3	4	5
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



See the highlighted item in red – the sum of yellow highlighted items in corresponding row and column is 23 + 23 = 46 - less than the original value infinity. So the highlighted item in red will be replaced by 46.

NPTEL

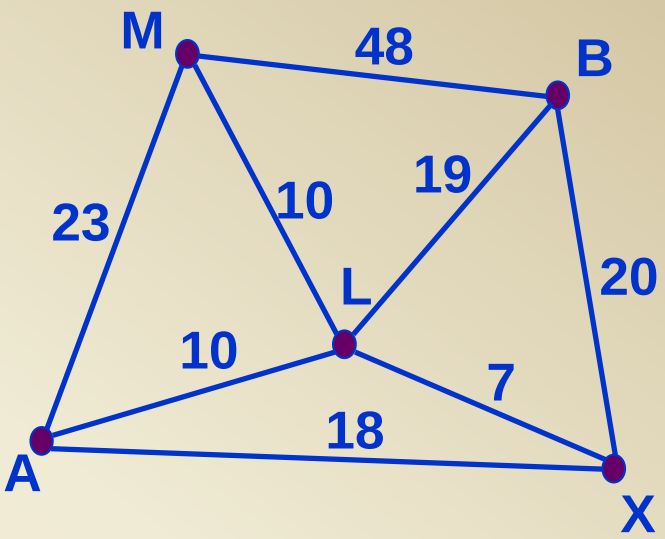
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	∞
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	2	3	4	5
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



See the highlighted item in red – the sum of yellow highlighted items in corresponding row and column is 23 + 10 = 33 – more than the original value 10!
So the highlighted item in red will remain at 10.

NPTEL

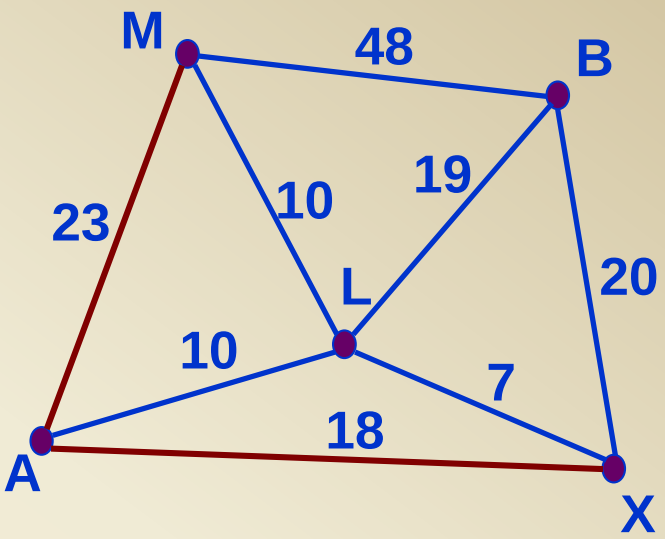
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	∞
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	2	3	4	5
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



Using similar procedure, we complete the 2nd row. See that 2 highlighted items are now changed in D(0) – to 46 and 41. Since changes are through Node 1 (i.e. A) Corresponding R(0) values are also changed to 1.

NPTEL

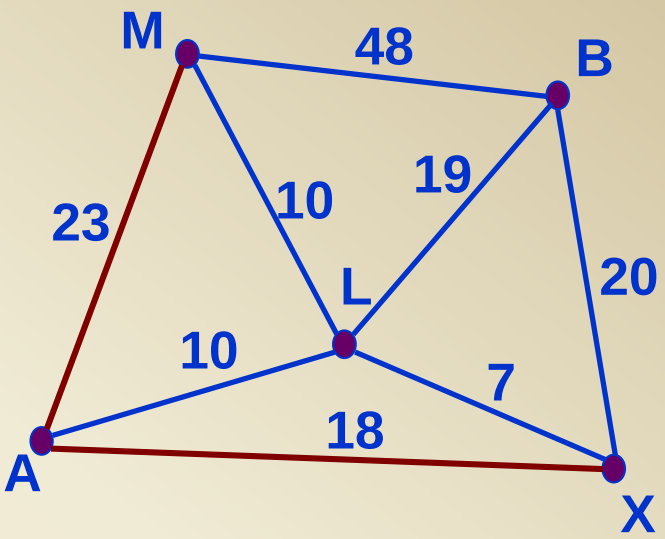
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	∞	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	3	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



Using similar procedure, we complete the 3nd row. See that 1 highlighted item is now changed in D(0) – to 20.
Since change is through Node 1 (i.e. A) Corresponding R(0) value is also changed to 1.

NPTEL

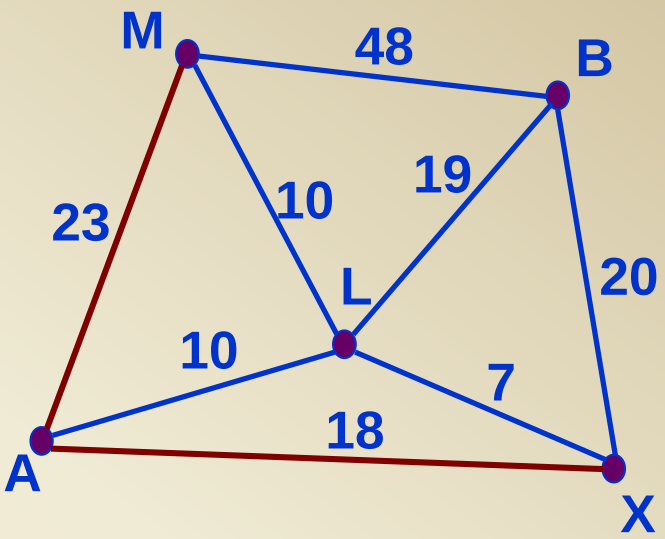
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	20	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	1	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



Nothing changes in the 4th row. Because for none of the items, we have a lower sum of the corresponding 1st row and column highlighted values (in yellow).
So no more changes in D(0) or R(0).

NPTEL

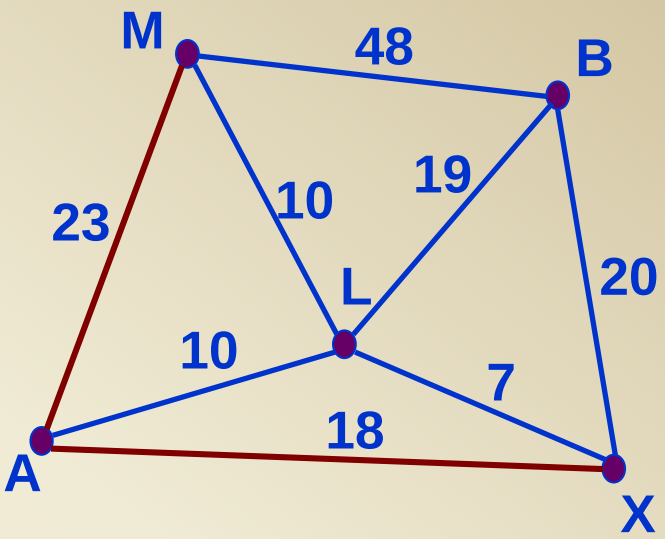
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	20	19	7
B	∞	48	19	∞	20
X	18	∞	7	20	∞

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	1	4	5
B	1	2	3	4	5
X	1	2	3	4	5

R(0) Route Matrix

Floyd's Algorithm



Using similar procedure, we complete the 5th row. See that 2 highlighted items are now changed in D(0) – to 41 and 36. Since change is through Node 1 (i.e. A) Corresponding R(0) values are also changed to 1.

NPTEL

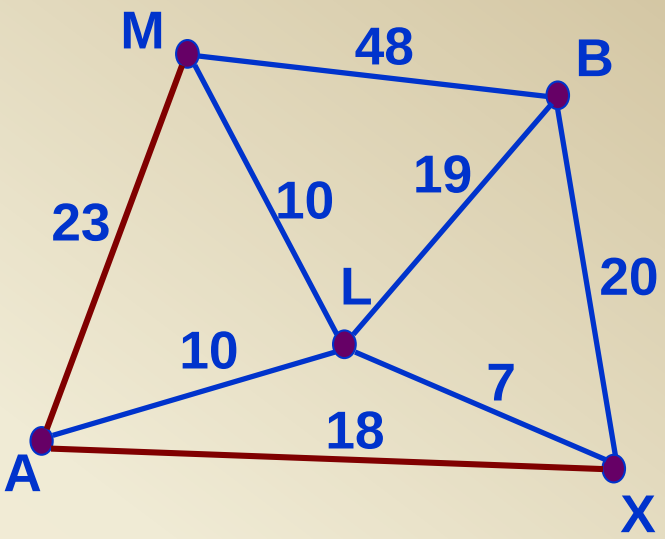
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	20	19	7
B	∞	48	19	∞	20
X	18	41	7	20	36

D(0) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	1	4	5
B	1	2	3	4	5
X	1	1	3	4	1

R(0) Route Matrix

Floyd's Algorithm



We now have completed iterations with point 1 (i.e. A). Similar iterations are needed for the rest of the points.

We therefore have now obtained D(1) and R(1).

NPTEL

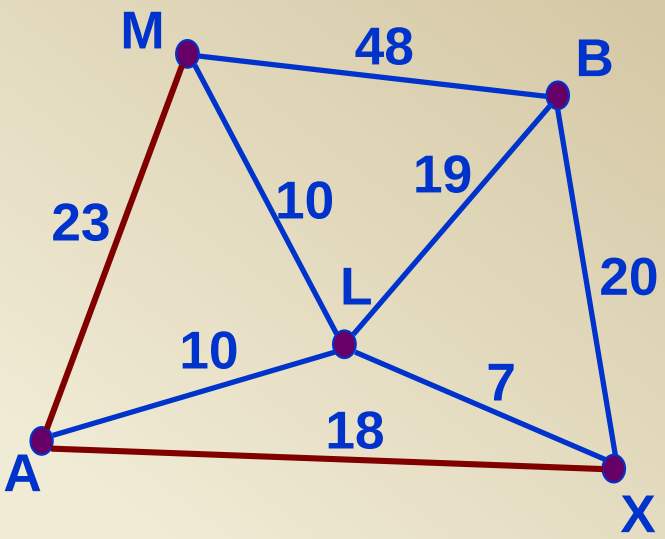
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	20	19	7
B	∞	48	19	∞	20
X	18	41	7	20	36

D(1) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	1	4	5
B	1	2	3	4	5
X	1	1	3	4	1

R(1) Route Matrix

Floyd's Algorithm



We highlight the 2nd column and row of D(1) and 2nd column of R(1); and compare all other items with the sum of the items highlighted in the same row and column. If the sum is less than the item then it should be replaced with the sum.

NPTEL

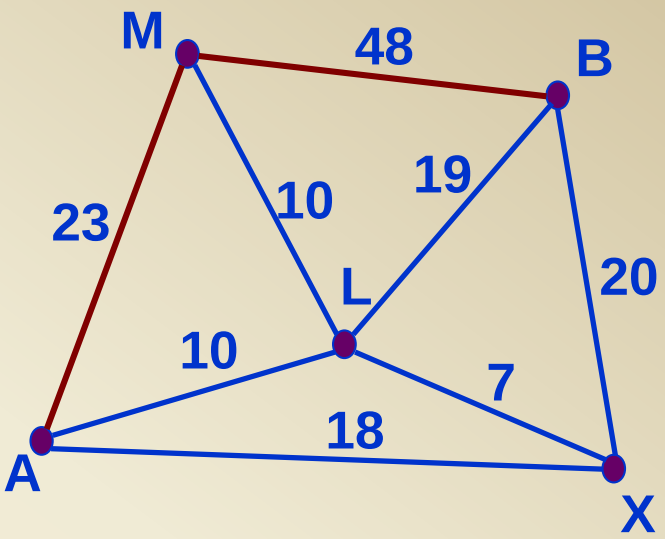
	A	M	L	B	X
A	∞	23	10	∞	18
M	23	46	10	48	41
L	10	10	20	19	7
B	∞	48	19	∞	20
X	18	41	7	20	36

D(1) Distance Matrix

	A	M	L	B	X
A	1	2	3	4	5
M	1	1	3	4	1
L	1	2	1	4	5
B	1	2	3	4	5
X	1	1	3	4	1

R(1) Route Matrix

Floyd's Algorithm



Using the procedure given, we find that 4 values are changed as highlighted in red to 46, 71, 71, and 96 respectively in D(1). Similarly, Corresponding values are changed in R(1) to 2 – because changes are through point 2 (i.e. M).

NPTEL

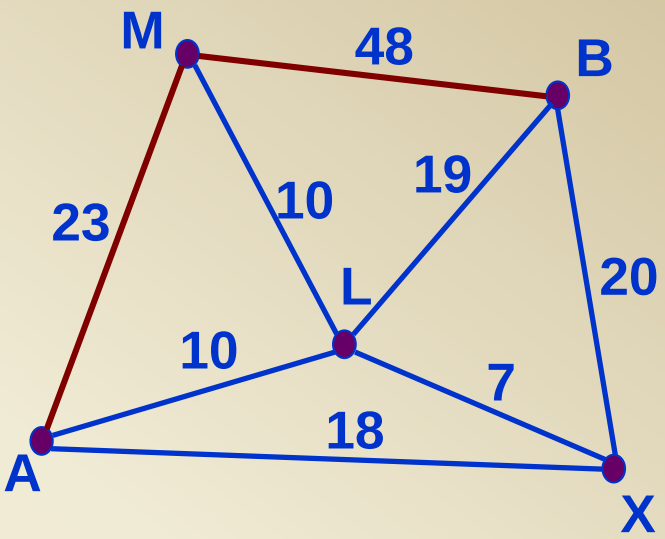
	A	M	L	B	X
A	46	23	10	71	18
M	23	46	10	48	41
L	10	10	20	19	7
B	71	48	19	96	20
X	18	41	7	20	36

D(1) Distance Matrix

	A	M	L	B	X
A	2	2	3	2	5
M	1	1	3	4	1
L	1	2	1	4	5
B	2	2	3	2	5
X	1	1	3	4	1

R(1) Route Matrix

Floyd's Algorithm



We now have completed iterations with point 2 (i.e. M). Similar iterations are needed for the rest of the points.

We therefore have now obtained D(2) and R(2).

NPTEL

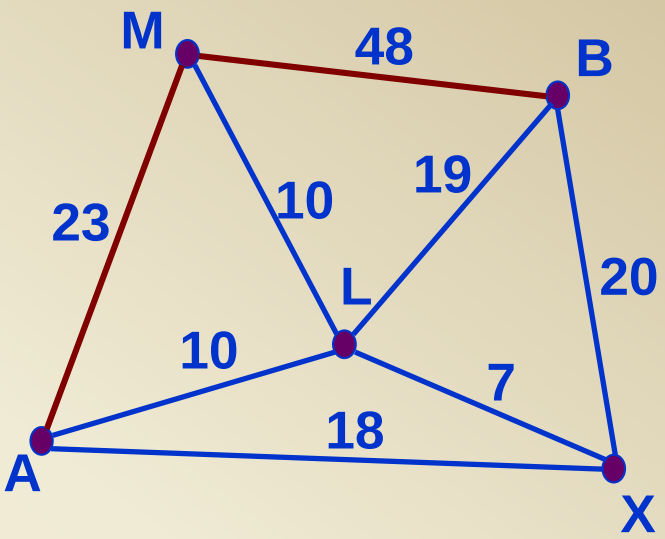
	A	M	L	B	X
A	46	23	10	71	18
M	23	46	10	48	41
L	10	10	20	19	7
B	71	48	19	96	20
X	18	41	7	20	36

D(2) Distance Matrix

	A	M	L	B	X
A	2	2	3	2	5
M	1	1	3	4	1
L	1	2	1	4	5
B	2	2	3	2	5
X	1	1	3	4	1

R(2) Route Matrix

Floyd's Algorithm



We highlight the 3rd column and row of D(2) and 3rd column of R(2); and compare all other items with the sum of the items highlighted in the same row and column. If the sum is less than the item then it should be replaced with the sum.

NPTEL

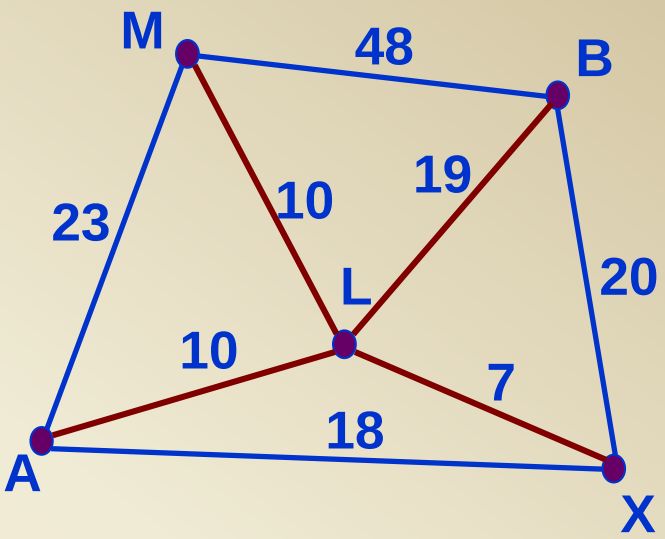
	A	M	L	B	X
A	46	23	10	71	18
M	23	46	10	48	41
L	10	10	20	19	7
B	71	48	19	96	20
X	18	41	7	20	36

D(2) Distance Matrix

	A	M	L	B	X
A	2	2	3	2	5
M	1	1	3	4	1
L	1	2	1	4	5
B	2	2	3	2	5
X	1	1	3	4	1

R(2) Route Matrix

Floyd's Algorithm



Using the procedure given, we find that 14 values are changed as highlighted in red to in D(2).

Similarly, Corresponding values are changed in R(2) to 3 – because changes are through point 3 (i.e. L).

NPTEL

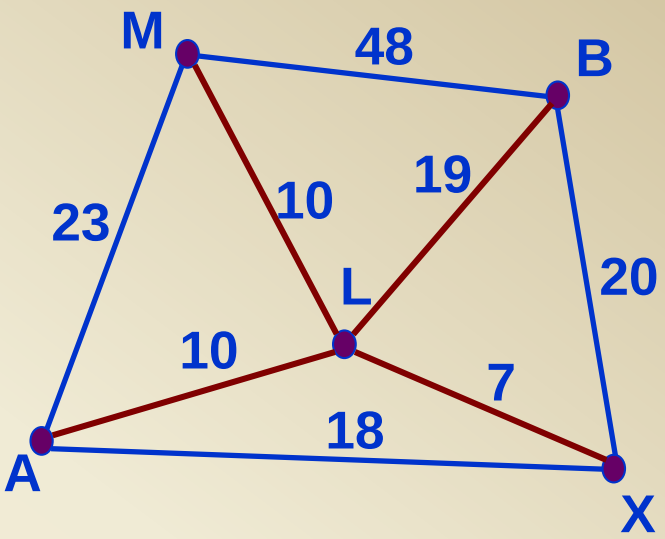
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	20	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(2) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	1	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(2) Route Matrix

Floyd's Algorithm



We now have completed iterations with point 3 (i.e. L). Similar iterations are needed for the rest of the points.

We therefore have now obtained D(3) and R(3).

NPTEL

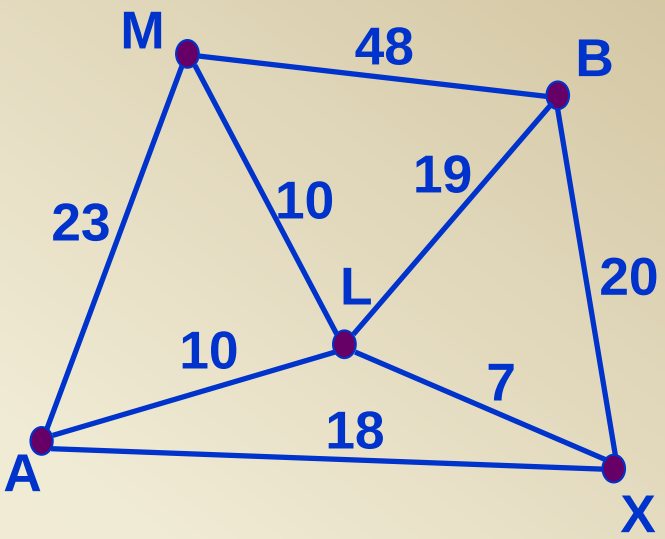
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	20	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(3) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	1	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(3) Route Matrix

Floyd's Algorithm



We highlight the 4th column and row of D(3) and 4th column of R(3); and compare all other items with the sum of the items highlighted in the same row and column. If the sum is less than the item then it should be replaced with the sum.

NPTEL

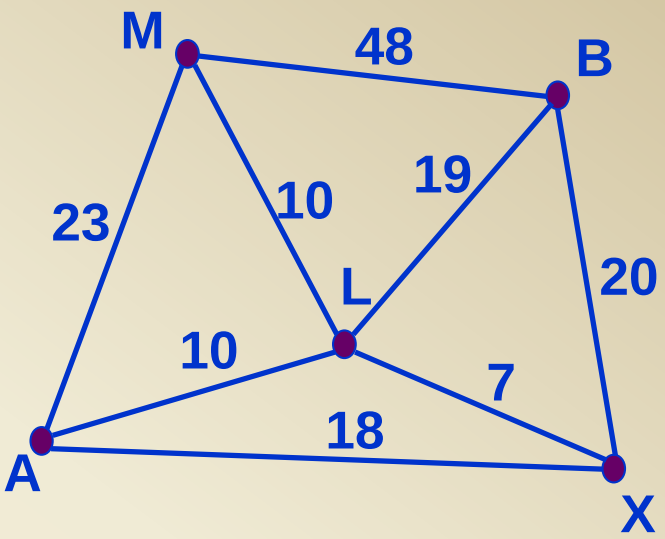
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	20	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(3) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	1	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(3) Route Matrix

Floyd's Algorithm



As no items could be changed, we have now completed iterations with point 4 (i.e. B). Similar iterations are needed for the final point X. We therefore have now obtained D(4) and R(4).

NPTEL

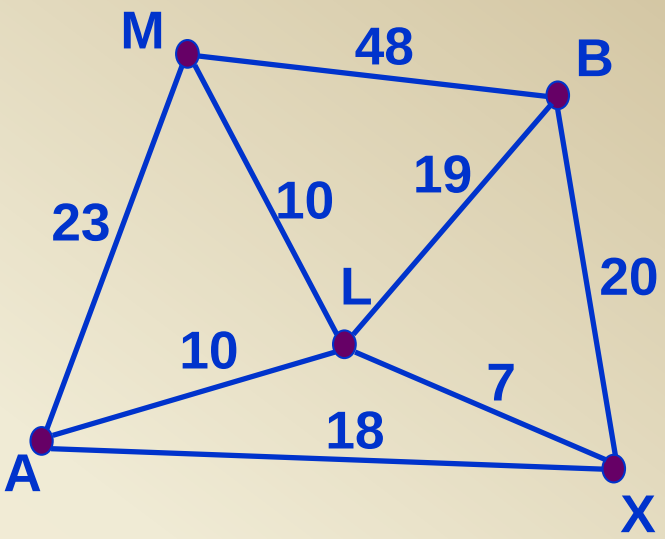
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	20	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(4) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	1	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(4) Route Matrix

Floyd's Algorithm



We highlight the 5th column and row of D(4) and 5th column of R(4); and compare all other items with the sum of the items highlighted in the same row and column. If the sum is less than the item then it should be replaced with the sum.

NPTEL

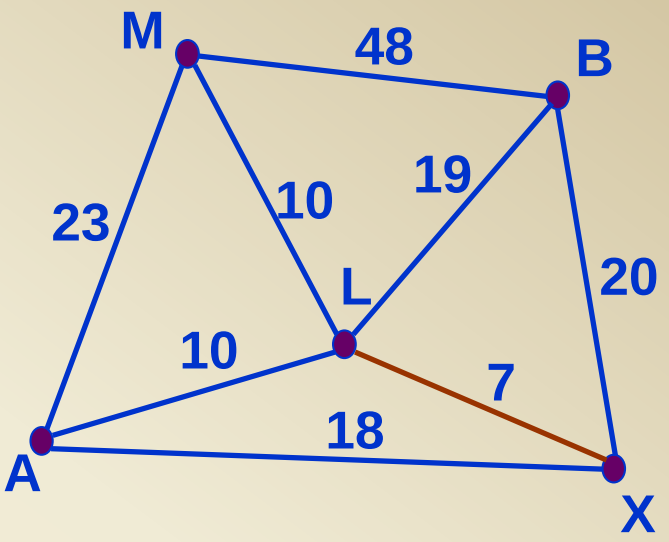
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	20	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(4) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	1	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(4) Route Matrix

Floyd's Algorithm



Using the procedure given, we find that 1 value is changed as highlighted in red to 14 as shown in D(4).
Similarly, Corresponding value is changed in R(4) to 5 – because changes are through point 5 (i.e. X).

NPTEL

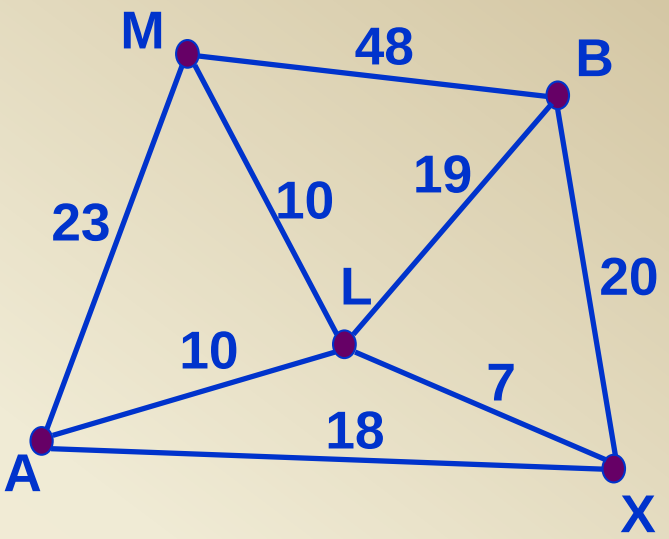
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	14	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(4) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	5	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(4) Route Matrix

Floyd's Algorithm



Now the final matrices D(5) and R(5) are obtained. These matrices will give shortest distances from any point to any other point. For example, between A to B, the shortest distance is 29 (See D(5)) and the route is A to L to B (See R(5)).

NPTEL

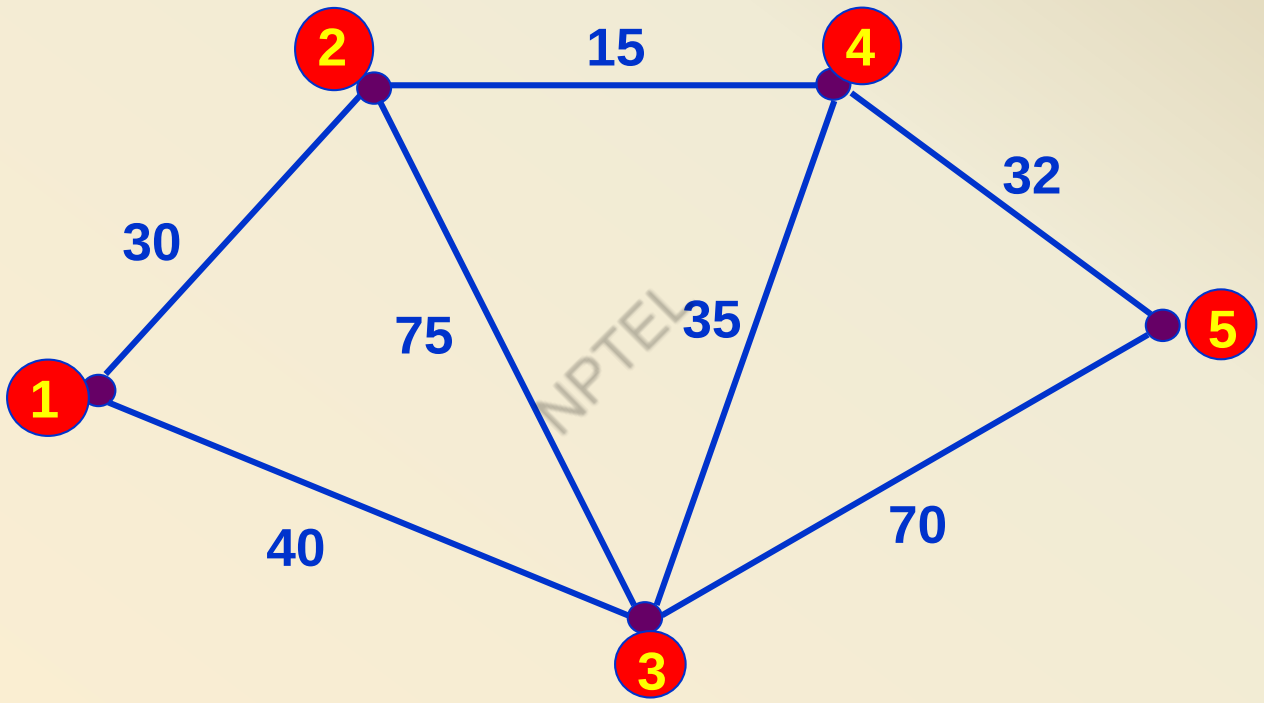
	A	M	L	B	X
A	20	20	10	29	17
M	20	20	10	29	17
L	10	10	14	19	7
B	29	29	19	38	20
X	17	17	7	20	14

D(5) Distance Matrix

	A	M	L	B	X
A	3	3	3	3	3
M	3	3	3	3	3
L	1	2	5	4	5
B	3	3	3	3	5
X	3	3	3	4	3

R(5) Route Matrix

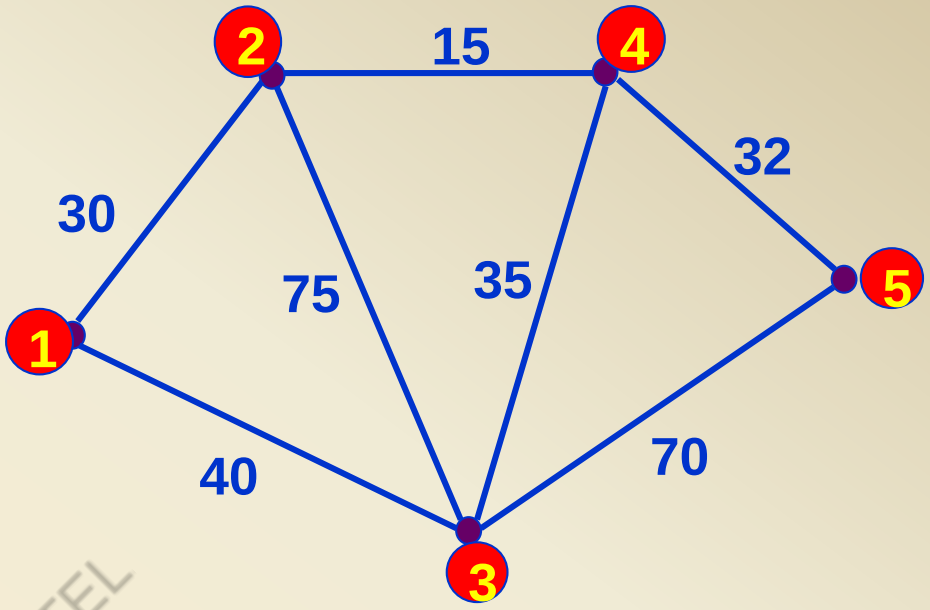
Exercise for Floyd's Algorithm



NPTEL

Find the Shortest Path from any node to any other node.

Solution to the Exercise on Floyd's Algorithm



NPTEL

	1	2	3	4	5
1	60	30	40	45	77
2	30	30	50	15	47
3	40	50	70	35	67
4	45	15	35	30	32
5	77	47	67	32	64

D(5)

	1	2	3	4	5
1	2	2	3	2	2
2	1	4	4	4	4
3	1	4	4	4	4
4	2	2	3	2	5
5	4	4	4	4	4

R(5)

THANK YOU